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Online reinforcement learning of state representation in recurrent network: the power of random feedback and biological constraints

Takayuki Tsurumi, Ayaka Kato, Arvind Kumar, Kenji Morita

Physical and Health Education, Graduate School of Education, The University of Tokyo, Tokyo, Japan • Theoretical Sciences Visiting Program, Okinawa Institute of Science and Technology, Okinawa, Japan • Department of Psychiatry, Icahn School of Medicine at Mount Sinai, New York, United States • Division of Computational Science and Technology, School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology, Stockholm, Sweden • International Research Center for Neurointelligence (WPI-IRCN), The University of Tokyo, Tokyo, Japan

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This work models reinforcement-learning experiments using a recurrent neural network. It examines if the detailed credit assignment necessary for back-propagation through time can be replaced with random feedback. In this **important** study the authors show that it yields a satisfactory approximation and the evidence to support that it holds within relatively simple tasks is **solid**.

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Abstract

Representation of external and internal states in the brain plays a critical role in enabling suitable behavior. Recent studies suggest that state representation and state value can be simultaneously learnt through Temporal-Difference-Reinforcement-Learning (TDRL) and Backpropagation-Through-Time (BPTT) in recurrent neural networks (RNNs) and their readout. However, neural implementation of such learning remains unclear as BPTT requires offline update using transported downstream weights, which is suggested to be biologically implausible. We demonstrate that simple online training of RNNs using TD reward prediction error and random feedback, without additional memory or eligibility trace, can still learn the structure of tasks with cue-reward delay and timing variability. This is because TD learning itself is a solution for temporal credit assignment, and feedback alignment, a mechanism originally proposed for supervised learning, enables gradient approximation without weight transport. Furthermore, we show that biologically constraining downstream weights and random feedback to be non-negative not only preserves learning but may even enhance it because the non-negative constraint ensures loose alignment - allowing the downstream and feedback weights to roughly align from the beginning. These results provide insights into the

neural mechanisms underlying the learning of state representation and value, highlighting the potential of random feedback and biological constraints.

Introduction

Multiple lines of studies have suggested that Temporal-Difference-Reinforcement-Leaning (TDRL) is implemented in the cortico-basal ganglia-dopamine (DA) circuits where DA encodes TD reward-prediction-error (RPE) [1–6] and DA-dependent plasticity of cortico-striatal synapses corresponds to TD-RPE-dependent update of state/action values [7–9]. Traditionally, TDRL in the cortico-basal ganglia-DA circuits was considered to serve only for relatively simple behavior. However, subsequent studies suggested that more sophisticated, apparently goal-directed/model-based behavior can also be achieved by TDRL if states are appropriately represented [10–12] and that DA signals indeed reflect model-based predictions [13, 14]. Conversely, impairments in state representation may relate to behavioral or mental-health problems [15–19]. Early modeling studies treated state representations appropriate to the situation/task as given ('handcrafted' by the authors), but representation itself should be learnt in the brain [20–25]. Recently it was shown that appropriate state representation can be learnt through TDRL in a recurrent neural network (RNN) by minimizing squared TD value-error without explicit target [12, 26], while state value can be simultaneously learnt in connections downstream of the RNN.

However, whether such a learning -referred to as the value-RNN [26], can be implemented in the brain remains unclear. This is because the value-RNN [26] used the Backpropagation-Through-Time (BPTT) [27], which applies gradient-descent error-'backpropagation' (hereafter referred to as backprop) [28, 29] to a temporally unfolded RNNs. BPTT has been argued to be biologically implausible mainly due to problems with feedback and causality. Regarding feedback, updating upstream connections requires feedback whose weights are transported from downstream forward connections, but such weight transportation is difficult to implement biologically [30, 31]. In the case of the value-RNN, if the state-representing RNN and the value-encoding readout are implemented by the intra-cortical circuit and the striatal neurons, respectively, as generally assumed [1, 32, 33], this weight transportation means that the update (plasticity) rule for intra-cortical connections involves the downstream cortico-striatal synaptic strengths, which are not accessible from the cortex. Regarding the problem of causality in BPTT, the error needs to be incrementally accumulated in the temporally backward order, but such an acausal offline update is biologically implausible [34, 35].

Recently, a potential solution for the feedback problem of backprop has been proposed [36] (see also [37–44] for other approaches). Specifically, in supervised learning of feed-forward networks, it was shown that when the transported downstream weights used for updating upstream connections were replaced with fixed random strengths, comparable learning performance was still achieved [36]. This was suggested to be because the information of the random strengths transferred to the upstream connections and then to the downstream feed-forward connections so that these feed-forward connections became aligned to the random feedback and thereby the random feedback could work similarly to the feedback with transported downstream weights in backprop. This mechanism was named the 'feedback alignment' (FA)[36], and was subsequently shown to work also in online supervised learning of RNN [34] and proposed to be 'neurally' implemented [45] (in a different way from the present study as we discuss in the Discussion).

The value-RNN [12, 26] differs from supervised learning considered in these previous feedback-alignment studies in two ways: i) it is TD learning, i.e., it approximates the true error by the TD-RPE because the true error, or true state value, is unknown, and ii) it uses a scalar error (TD-RPE) rather than a vector error. Scalar reward-based online learning of RNN with random feedback was actually shown to work in a different study [35] (their Supplementary Figure 5), but TD-RPE was not introduced in that setup, while the same study also examined another setup with TD-RPE but result with random feedback was not shown for this latter setup (their Figures 4 and 5). Therefore, it was nontrivial whether the value-RNN could be modified to incorporate online update using random feedback. Here we demonstrate that such a modified value-RNN could still work and provide a mechanistic insight into how it works.

Next we address other biological-plausibility issues with FA-based rule. Specifically, we imposed biological constraints that the downstream (cortico-striatal) weights and the fixed random feedback, as well as the activities of neurons in the RNN, were all non-negative. Moreover, we also remedied the non-monotonic dependence of the update of RNN connection-strength on post-synaptic neural activity. We found that the non-negative constraint appeared to aid, rather than degrade, the learning by ensuring that the downstream weights and the fixed random feedback were loosely aligned from the beginning. These results suggest how learning of state representation and value could be implemented via DA-dependent synaptic plasticity in cortical and striatal circuits, where DA encodes TD-RPE.

Results

1. Online value-RNN with fixed random feedback

We considered an online value-RNN in the cortico-basal ganglia circuits (**Fig. 1**). In this model a cortical region/population represents information of sensory observation ($\boldsymbol{o}$) and send it to another cortical region/population which estimates a state ($\boldsymbol{x}$) given the sensory inputs. We approximate this cortical population an RNN (number of RNN units was varied between 5 and 40). Neurons in the RNN learn to represent states by updating the strengths of recurrent connections **A** and feed-forward connections **B**. The activity of a population of striatal neurons that receive inputs from the RNN is supposed to learn to represent the state values (v), by learning the weights ($\boldsymbol{w}$) of cortico-striatal connections. DA neurons in the ventral tegmental area (VTA) receive information about the value and reward (r) from the striatum (both direct and indirect pathways) and other structures, and the activity of the DA neurons, as well as released DA, represents TD-RPE (δ). The TD-RPE-representing DA is released in the striatum and also in the cortical RNN through mesocorticolimbic projections, and used for modifying **A**, **B** and $\boldsymbol{w}$ (**Fig. 1**).

The original value-RNN [12], [26] adopted BPTT [27] as an update rule for the connections onto the RNN (**A** and **B**), which requires the (gradually changing) value weights ($\boldsymbol{w}$), but this is biologically implausible because the corticostriatal synaptic weights are not available in the cortex as mentioned above. Therefore, we examined a model (agent) in which these weights were replaced with fixed random strengths ($\boldsymbol{c}$), in comparison with a model that used these weights. Because the temporally acausal offline update used in BPTT is also biologically implausible, we used an online learning rule, which considers only the influence of the recurrent weights at the previous time step (see the Methods for details and equations). We refer to these models (agents) as the online value-RNN or oVRNN.

We assumed that a single RNN unit corresponds to a small population of neurons that intrinsically share inputs and outputs, for genetic or developmental reasons, and the activity of each unit represents the (relative) firing rate of the population. Cortical population activity is suggested to be sustained not only by fast synaptic transmission and spiking but also, even predominantly, by slower synaptic neurochemical dynamics [46] such as short-term facilitation, whose time constant can be around 500 milliseconds [47]. Therefore, we assumed that single time-step of our rate-based (rather than spike-based) model corresponds to 500 milliseconds.

In each simulation, the recurrent (**A**) and feed-forward connection (**B**) weights onto the RNN units were initialized to pseudo standard normal random numbers. As a negative control, we also conducted simulations in which these connections were not updated from initial values, referring

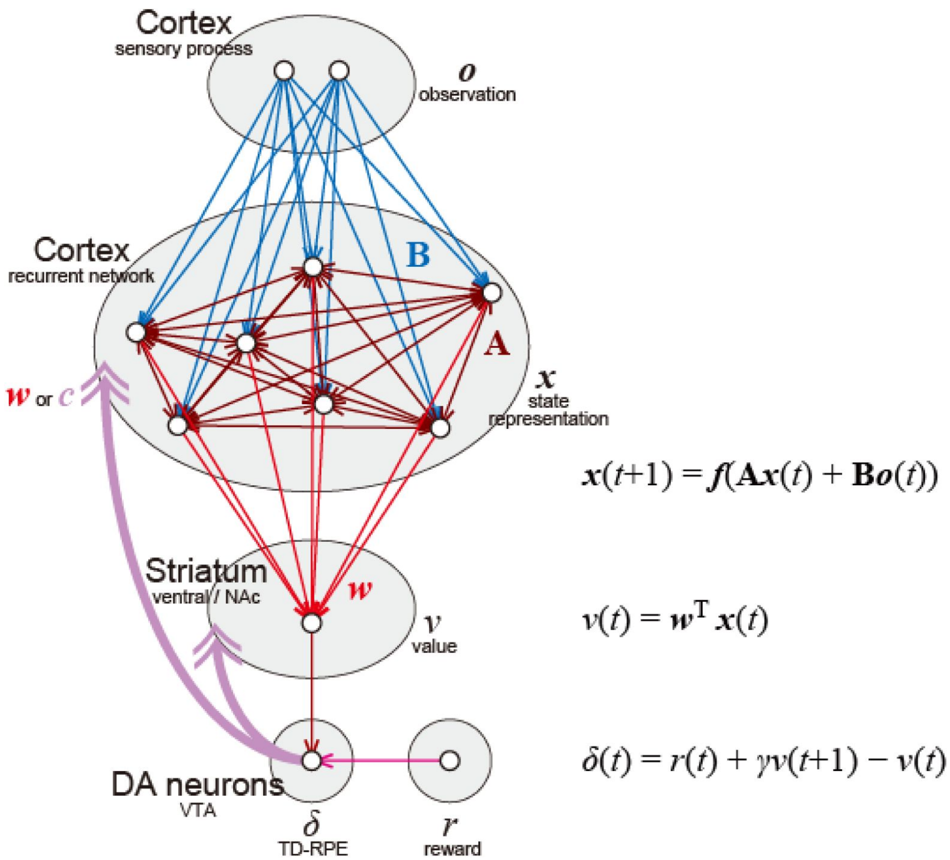

Figure 1

Implementation of the online value-RNN in the cortico-basal ganglia-DA circuits.

to as the case with “untrained (fixed) RNN”. Notably, the value weights $\boldsymbol{w}$ (i.e., connection weights from the RNN to the striatal value unit) were still trained in the models with untrained RNN. The oVRNN models, and the model with untrained RNN, were continuously trained across trials in each task, because we considered that it was ecologically more plausible than episodic training of separate trials.

2.1 Simulation of a Pavlovian cue-reward association task with variable inter-trial intervals

First we took a small RNN with 7 units to represent state in the cortex, and simulated a Pavlovian cue-reward association task, in which a cue was followed by a reward three time-steps later, and inter-trial interval (i.e., reward to next cue) was randomly chosen from 4, 5, 6, or 7 time-steps (**Fig. 2A**). Given that single time-step corresponds to 500 milliseconds as mentioned above, three time-steps from cue to reward correspond to 1.5 sec, which matches the delay in the conditioning task used by Schultz et al. [2]. In this task, states after receiving of cue information can be defined by time-steps from the cue, and the state values of these states can be estimated by calculating the expected cumulative discounted future rewards [48] through simulations; we refer to them as “estimated true state values” (**Fig. 2B**, black line). Expected TD-RPE can be calculated from these estimated true values (**Fig. 2B**, red line).

First, for comparison, we examined traditional TD-RL agent with punctate state representation (without using the RNN), in which each state (time-step from a cue) was represented in a punctate manner, i.e., by a one-hot vector such as (1, 0, ..., 0), (0, 1, ..., 0), and so on. We examined two cases: one in which training was done in an episodic manner without continuation between trials (i.e., the value of the last state in a trial was not updated by TD-RPE upon entering the next trial) and the other in which training was done continuously across trials, as in the cases of agents using the RNN. The former agent developed positive values between cue and reward, and abrupt TD-RPE upon cue (**Fig. 2C**), whereas the latter agent developed positive values also for states in the inter-trial interval (**Fig. 2D**), looking similar to the true values (**Fig. 2B**).

We then examined our oVRNN agents, with backprop-type transported downstream weights (oVRNNbp: **Fig. 2E**) or with fixed random feedback (oVRNNrf: **Fig. 2F**), in comparison with the agent with untrained fixed RNN (**Fig. 2G**). As shown in the figures, oVRNNbp successfully learned the values of states between cue and reward, and oVRNNrf also learned these values, although to a somewhat smaller degree on average. On the other hand, the agent with untrained RNN developed the smallest state values on average among the three agents. This inferiority of untrained RNN may sound odd because there were only four states from cue to reward while random RNN with enough units is expected to be able to represent many different states (c.f., [49]) and the effectiveness of training of only the readout weights has been shown in reservoir computing studies [50]–[53]. However, there was a difficulty stemming from the continuous training across trials (rather than episodic training of separate trials): the activity of untrained RNN upon cue presentation generally differed from trial to trial, and so it is non-trivial that cue presentation in different trials should be regarded as the same single state, even if it could eventually be dealt with at the readout level if the number of units increases.

The results above indicate that value-RNN could be trained online by fixed random feedback at least to a certain extent, although somewhat less effectively than by backprop-type feedback. Results of individual simulations shown in **Fig. 2H,I** indicate that state values developed in oVRNNrf were largely comparable to those developed in oVRNNbp once they were successfully learned but the success rate was smaller than oVRNNbp while still larger than the untrained RNN.

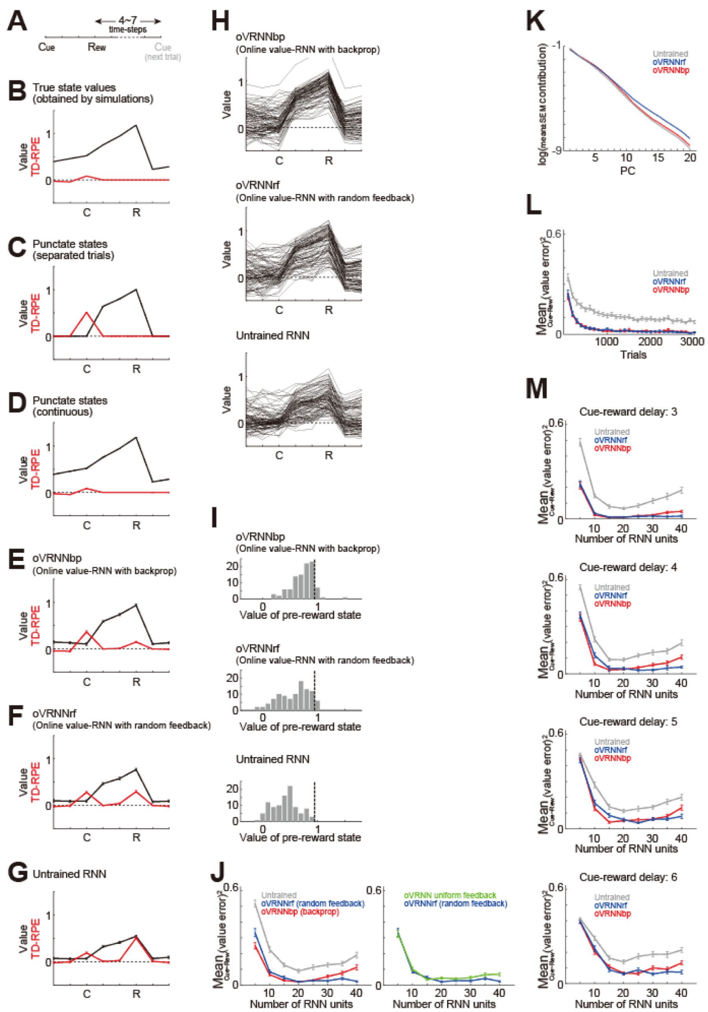

Figure 2

Simulation of a Pavlovian cue-reward association task.

(A) Simulated task with variable inter-trial intervals. **(B)** Black line: Estimated true values of states/timings through simulations according to the definition of state value, i.e., expected cumulative discounted future rewards, taking into account the effect of probabilistic inter-trial interval (ITI). Red line: TD-RPEs calculated from the estimated true state/timing values. **(C-G)** State values (black lines) and TD-RPEs (red lines) at 1000-th trial, averaged across 100 simulations (error-bars indicating mean ± SEM across simulations; same applied to the followings unless otherwise mentioned), in different types of agent: (C) TD-RL agent having punctate state representation and state values without continuation between trials (i.e., the value of the last state in a trial was not updated by TD-RPE upon entering the next trial); (D) TD-RL agent having punctate state representation and continuously updated state values across trials; (E) Online value-RNN with backprop (oVRNNbp). The number of RNN units was 7 (same applied to (F,G); (F) Online value-RNN with fixed random feedback (oVRNNrf); (G) Agent with untrained RNN. **(H)** State values at 1000-th trial in individual simulations of oVRNNbp (top), oVRNNrf (middle), and untrained RNN (bottom). **(I)** Histograms of the value of the pre-reward state (i.e., the state one-time step before the reward state) at 1000-th trial in individual simulations of the three models. The vertical black dashed lines indicate the true value of the pre-reward state (estimated through simulations). **(J)** *Left*: Mean of the squares of differences between the state values developed by each agent and the estimated true state values between cue and reward (referred to as the mean squared value-error) at 1000-th trial in oVRNNbp (red line), oVRNNrf (blue line), and the model with untrained RNN (gray line) when the number of RNN units (n) was varied from 5 to 40. Learning rate for value weights was normalized by dividing by n/7 (same applied to the followings unless otherwise mentioned). *Right*: Mean squared value-error in oVRNNrf (blue line: same data as in the left panel) and oVRNN with uniform feedback (green line). **(K)** Log contribution ratios of the principal components of the time series (for 1000 trials) of RNN activities in each model with 20 RNN units. **(L)** Mean squared value-error in each model with 20 RNN units across trials. **(M)** Mean squared value-error in each model at 3000-th trial in the cases where the cue-reward delay was 3, 4, 5, or 6 time-steps (top to bottom panels).

2.2 Systematic simulations and analyses

Next, we tested whether learning performance of oVRNNbp, oVRNNrf, and the agent with untrained fixed RNN depends on the number of RNN units (n). For a valid comparison with the previously shown cases with 7 RNN units, the learning rate for the value weights was normalized by dividing by $n/7$. Learning performance was measured by the mean of squares of differences between the state values developed by each of these three types of agents and the estimated true state values (**Fig. 2B**) between cue and reward at 1000-th trial. As shown in the left panel of **Fig. 2J**, on average across simulations, oVRNNbp and oVRNNrf exhibited largely comparable performance and always outperformed the untrained RNN ($p < 0.00022$ in Wilcoxon rank sum test for oVRNNbp or oVRNNrf vs untrained for each number of RNN units), although oVRNNbp somewhat outperformed or underperformed oVRNNrf when the number of RNN units was small (≤10 ($p < 0.049$)) or large (≥25 ($p < 0.045$)), respectively. As the number of RNN units increased from 5 to 15 or 20, all these three agents improved their performance. Additional increase of RNN units did not largely change the mean performance in oVRNNrf, while moderately decreased it in oVRNNbp and untrained RNN. The green line in **Fig. 2J-right** shows the performance of a special case where the random feedback in oVRNNrf was fixed to the direction of $(1, 1, ..., 1)^T$ (i.e., uniform feedback) with a random coefficient, which was largely comparable to, but somewhat worse than, that for the general oVRNNrf (blue line).

In order to examine the dimensionality of RNN dynamics, we conducted principal component analysis (PCA) of the time series (for 1000 trials) of RNN activities and calculated the contribution ratios of PCs in the cases of oVRNNbp, oVRNNrf, and untrained RNN with 20 RNN units. **Figure 2K** shows a log of contribution ratios of 20 PCs in each case. Compared with the case of untrained RNN, in oVRNNbp and oVRNNrf, initial component(s) had smaller contributions (PC1 (t-test $p = 0.00018$ in oVRNNbp; $p = 0.0058$ in oVRNNrf) and PC2 ($p = 0.080$ in oVRNNbp; $p = 0.0026$ in oVRNNrf)) while later components had larger contributions (PC3~10,15~20 $p < 0.041$ in oVRNNbp; PC5~20 $p < 0.0017$ in oVRNNrf) on average, and this is considered to underlie their superior learning performance. We noticed that late components had larger contributions in oVRNNrf than in oVRNNbp, although these two models with 20 RNN units were comparable in terms of cue~reward state values (**Fig. 2J-left**).

We examined how learning proceeded across trials in the models with 20 RNN units. As shown in **Fig. 2L**, learning became largely converged by 1000-th trial, although slight improvement continued afterward. We further examined the cases with longer cue-reward delays. As shown in **Fig. 2M**, as the delay increased, the mean squared error of state values (at 3000-th trial) increased, but the relative superiority of oVRNNbp and oVRNNrf over the model with untrained RNN remained to hold, except for cases with small number of RNN units (5) and long delay (5 or 6) ($p < 0.0025$ in Wilcoxon rank sum test for oVRNNbp or oVRNNrf vs untrained for each number of RNN units for each delay).

2.3 Occurrence of feedback alignment and an intuitive understanding of its mechanism

Next, we questioned how feedback alignment contributes to the learnability of oVRNNrf. To address this question we used an RNN with 7 units and examined whether the value weight vector $\boldsymbol{w}$ became aligned to the random feedback vector $\boldsymbol{c}$ in oVRNNrf, by looking at the changes in the angle between these two vectors across trials. As shown in **Fig. 3A**, this angle, averaged across simulations, decreased over trials, indicating that the value weight $\boldsymbol{w}$ indeed tended to become aligned to the random feedback $\boldsymbol{c}$. We then examined whether better alignment of $\boldsymbol{w}$ to $\boldsymbol{c}$ related to better development of state value by looking at the relation between the angle between $\boldsymbol{w}$ and $\boldsymbol{c}$ and the value of the pre-reward state at 1000-th trial. As shown in **Fig. 3B**, there was a negative correlation such that the smaller the angle (i.e., more aligned), the larger the state value (r =

−0.288, p = 0.00362), consistent with our expectation. These results indicate that the mechanism of feedback alignment, previously shown to work for supervised learning, also worked for TD learning of value weights and recurrent/feed-forward connections.

How did the feedback alignment mechanistically occur? We made an attempt to obtain an intuitive understanding. Assume that positive TD-RPE ($\delta(t) > 0$) is generated in a state, S (= $\boldsymbol{x}(t)$) in a trial. Because of the update rule for $\boldsymbol{w}$ ($\boldsymbol{w} \leftarrow \boldsymbol{w} + a\delta(t)\boldsymbol{x}(t)$) (**Eq. 1.9** in the Methods), $\boldsymbol{w}$ is updated in the direction of $\boldsymbol{x}(t)$. Next, what is the effect of updates of recurrent/feed-forward connections (**A** and **B**) on $\boldsymbol{x}$? For simplicity, here we consider the case where observation is null ($\boldsymbol{o}$ = **0**) and so $\boldsymbol{x}(t) = \boldsymbol{f}(\mathbf{A}\boldsymbol{x}(t-1))$ holds (but similar argument can be done in the case where observation is not null). If **A** is replaced with its updated one, it can be calculated that i-th element of $\mathbf{A}\boldsymbol{x}(t-1)$ will hypothetically change by c_i × (a positive value) (technical note: the value is $a\delta(t)\{\Sigma_j x_j(t-1)^2\}(0.5 + x_i(t))(0.5 - x_i(t))$ (c.f., **Eq. 1.10**) which is positive unless $x(t-1) = \mathbf{0}$), and therefore the vector $\mathbf{A}\boldsymbol{x}(t-1)$ as a whole will hypothetically change by a vector that is in a relatively close angle with $\boldsymbol{c}$, or more specifically, is in the same quadrant as (and thus within at maximum 90° from) $\boldsymbol{c}$ (for example, $[c_1\ c_2\ c_3]^\mathrm{T}$ and $[0.5c_1\ 1.2c_2\ 0.8c_3]^\mathrm{T}$). Then, because f is a monotonically increasing sigmoidal function, $\boldsymbol{x}(t) = \boldsymbol{f}(\mathbf{A}\boldsymbol{x}(t-1))$ will also hypothetically change by a vector that is in a relatively close angle with $\boldsymbol{c}$. This was indeed the case in our simulations as shown in **Fig. 3C**, which plotted the angle between the hypothetical change in $\boldsymbol{x}(t) = \boldsymbol{f}(\mathbf{A}\boldsymbol{x}(t-1),\mathbf{B}\boldsymbol{o}(t-1))$ in case **A** and **B** were replaced with their updated ones, multiplied with the sign of TD-RPE (sign($\delta(t)$)), and the fixed random feedback vector $\boldsymbol{c}$ across time-steps.

In this way, at state S where TD-RPE is positive, $\boldsymbol{w}$ is updated in the direction of $\boldsymbol{x}(t)$, and $\boldsymbol{x}(t)$ will hypothetically change by a vector that is in a relatively close angle with $\boldsymbol{c}$ if **A** is replaced with its updated one. Then, if the update of $\boldsymbol{w}$ and the hypothetical change in $\boldsymbol{x}(t)$ due to the update of **A** could be integrated, $\boldsymbol{w}$ would become aligned to $\boldsymbol{c}$ (if TD-RPE is instead negative, $\boldsymbol{w}$ is updated in the opposite direction of $\boldsymbol{x}(t)$, and $\boldsymbol{x}(t)$ will hypothetically change by a vector that is in a relatively close angle with $-\boldsymbol{c}$, and so the same story holds in the end).

There is, however, a caveat regarding how the update of $\boldsymbol{w}$ and the hypothetical change in $\boldsymbol{x}(t)$ can be integrated. Although technical, here we briefly describe the caveat and a possible solution for it. The updates of $\boldsymbol{w}$ and **A** use TD-RPE, which are calculated based on $v(t) = \boldsymbol{w}^\mathrm{T}\ \boldsymbol{x}(t)$ and $v(t+1) = \boldsymbol{w}^\mathrm{T}\ \boldsymbol{x}(t+1)$, and so $\boldsymbol{x}(t)$ and $\boldsymbol{x}(t+1)$ should already be determined beforehand. Therefore, the hypothetical change in $\boldsymbol{x}(t)$ due to the update of **A**, described in the above, does not actually occur (this was why we mentioned 'hypothetical') and thus cannot be integrated with the update of $\boldsymbol{w}$. Nevertheless, integration could still occur across successive trials, at least to a certain extent. Specifically, although TD-RPEs at S in successive trials would generally differ from each other, they would still tend to have the same sign, as was indeed the case in our simulations (**Fig. 3D**). Also, although the trajectories of RNN activity ($\boldsymbol{x}$) in successive trials would differ, we could expect a certain level of similarity because the RNN is entrained by observation-representing inputs, again as was indeed the case in our example simulation (**Fig. 3E**). Then, the hypothetical change in $\boldsymbol{x}(t)$ due to the update of **A**, considered above, could become a reality in the next trial, to a certain extent, and could thus be integrated into the update of $\boldsymbol{w}$, explaining the occurrence of feedback alignment.

2.4 Simulation of tasks with probabilistic structures of reward timing/existence

Previous work [54] examined the response of DA neurons in cue-reward association tasks in which reward timing was probabilistically determined (early in some trials but late in other trials). There were two tasks, which were largely similar but there was a key difference that reward was given in all the trials in one task whereas reward was omitted in some randomly determined trials in another task. Starkweather et al. [54] found that the DA response to later reward was smaller than the response to earlier reward in the former task, presumably reflecting

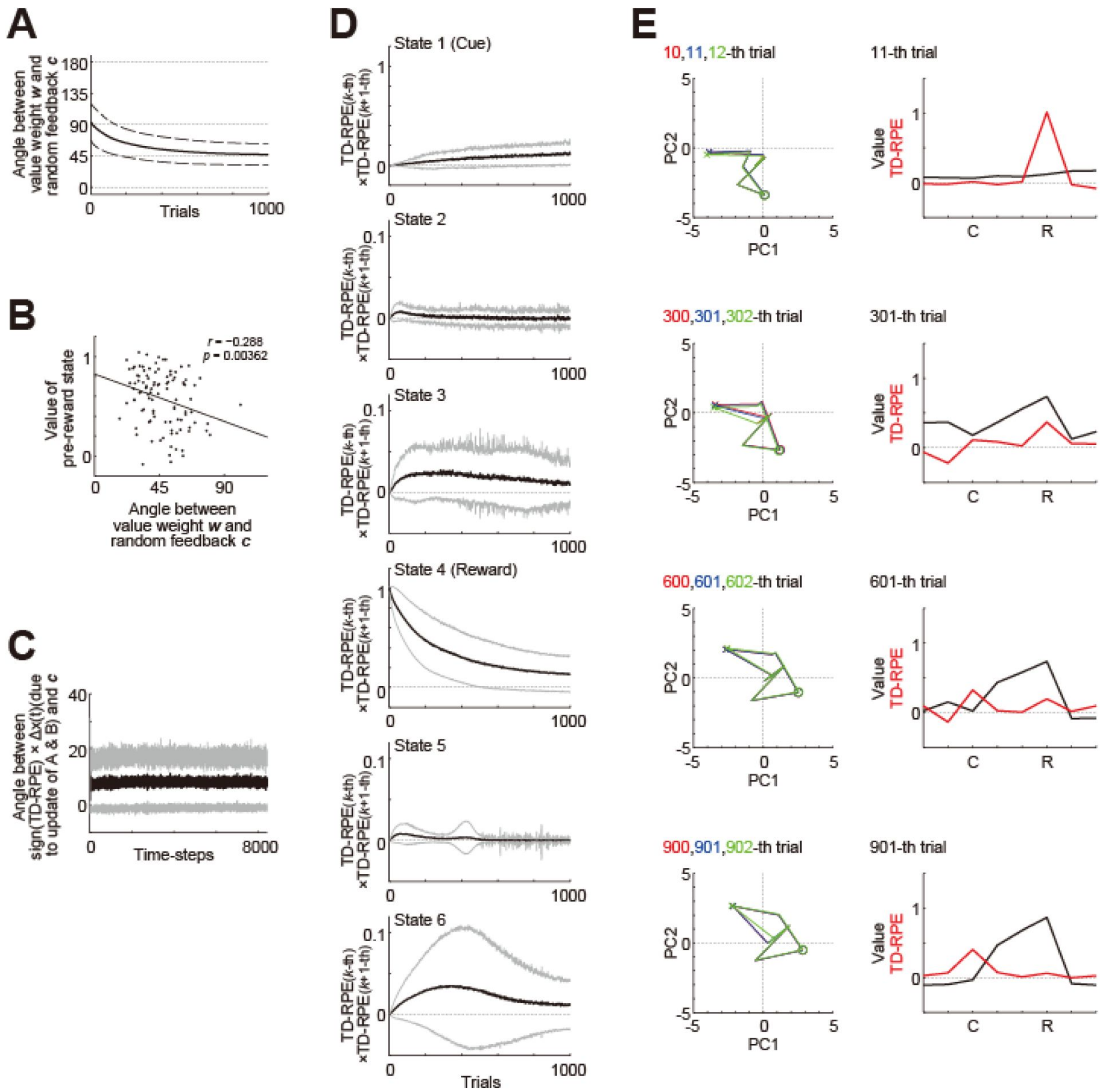

Figure 3

Occurrence of feedback alignment and an intuitive understanding of its mechanism.

(A) Over-trial changes in the angle between the value-weight vector ***w*** and the fixed random feedback vector ***c*** in the simulations of oVRNNrf (7 RNN units). The solid line and the dashed lines indicate the mean and ± SD across 100 simulations, respectively. **(B)** Negative correlation ($r = -0.288$, $p = 0.00362$) between the angle between ***w*** and ***c*** (horizontal axis) and the value of the pre-reward state (vertical axis) at 1000-th trial. The dots indicate the results of individual simulations, and the line indicates the regression line. **(C)** Angle between the hypothetical change in $\boldsymbol{x}(t) = \boldsymbol{f}(\mathbf{A}\boldsymbol{x}(t-1), \mathbf{B}\boldsymbol{o}(t-1))$ in case **A** and **B** were replaced with their updated ones, multiplied with the sign of TD-RPE (sign(δ(t))), and the fixed random feedback vector ***c*** across time-steps. The black thick line and the gray lines indicate the mean and ± SD across 100 simulations, respectively (same applied to (D)). **(D)** Multiplication of TD-RPEs in successive trials at individual states (top: cue, 4th from the top: reward). Positive or negative value indicates that TD-RPEs in successive trials have the same or different signs, respectively. **(E)** *Left*: RNN trajectories mapped onto the primary and secondary principal components (horizontal and vertical axes, respectively) in three successive trials (red, blue, and green lines (heavily overlapped)) at different phases in an example simulation (10th-12th, 300th-302th, 600th-602th, and 900th-902th trials from top to bottom). The crosses and circles indicate the cue and reward states, respectively. *Right*: State values (black lines) and TD-RPEs (red lines) at 11th, 301th, 601th, and 901th trial.

the animal's belief that delayed reward will surely come, but the opposite was the case in the latter task, presumably because the animal suspected that reward was omitted in that trial. Starkweather et al.[54] then showed that such response patterns could be explained if DA encoded TD-RPE under particular state representations that incorporated the probabilistic structures of the task (called the 'belief state'). In that study, such state representations were 'handcrafted' by the authors, but the subsequent work [26] showed that the original value-RNN with backprop (BPTT) could develop similar representations and reproduce the experimentally observed DA patterns.

In order to examine if our online value-RNN with fixed random feedback could also explain those experimental results, we simulated two tasks (**Fig. 4A**) that were qualitatively similar to (though simpler than) the two tasks examined in the experiments [54]. In our task 1, a cue was always followed by a reward either two or four time-steps later with equal probabilities. Task 2 was the same as task 1 except that reward was omitted with 40% probability. In task 1, if reward was not given at the early timing (i.e., two-steps later than cue), agent could predict that reward would be given at the late timing (i.e., four-steps later than cue), and thus TD-RPE upon reward at the late timing is expected to be smaller than TD-RPE upon reward at the early timing (if agent perfectly learned the task structure, TD-RPE upon reward at the late timing should be 0). By contrast, in task 2, if reward was not given at the early timing, it might indicate that reward would be given at the late timing but might instead indicate that reward would be omitted in that trial, and thus TD-RPE upon reward at the late timing is expected to exist and can even be larger than TD-RPE upon reward at the early timing.

In these tasks, states can be defined in the following way. There were two types of trials, with early or late reward, in task 1, and additionally one more type of trial, without reward, in task 2 (**Fig. 4Ba**, top). For each timing after receival of cue information in each of these trial types, its value can be estimated through simulations (**Fig. 4Ba**, bottom). Agent could not know the current trial type until receiving reward at the early timing or the late timing or receiving no reward at both timings. Until these timings, agent could have probabilistic belief about the current trial type, e.g., 50% in the trial with early reward and 50% in the trial with late reward (in task 1) or 30% in the trial with early reward, 30% in the trial with late reward, and 40% in the trial without reward (in task 2) (**Fig. 4Bb**). States of the timings after receival of cue information can be defined by incorporating these probabilistic beliefs at each timing (**Fig. 4Bc**, top). Their true values can be calculated by taking (mathematical) expected value of the estimated values of each timing in each trial type (**Fig. 4Bc**, bottom). Expected TD-RPE calculated from the estimated true values (**Fig. 4C**) exhibited features that matched the conjecture mentioned above: in task 1, TD-RPE upon reception of late reward, which was actually 0, was smaller than TD-RPE upon reception of early reward, whereas in task 2, TD-RPE upon reception of late reward was larger than TD-RPE upon reception of early reward (as indicated by the inequality signs on **Fig. 4C**).

As mentioned above, the previous work [54] has shown that VTA DA neurons exhibited similar activity patterns to the abovementioned TD-RPE patterns, and the subsequent work [26] has shown that the original value-RNN with backprop (BPTT) could reproduce such TD-RPE patterns. We examined how our oVRNNbp and oVRNNrf, (with 12 RNN units) behaved in our simulated two tasks. oVRNNbp developed the expected TD-RPE patterns, i.e., smaller TD-RPE upon late than early timing in task 1 but opposite pattern in task 2 (**Fig. 4F**), and oVRNNrf also developed such patterns although the effect size for task 1 was small (**Fig. 4G**). These results indicate that online value-RNN could learn the probabilistic structures of the tasks even with fixed random feedback. By contrast, agents with punctate state representation without or with continuous value update across trials (**Fig. 4D,E**), as well as agent with untrained fixed RNN (**Fig. 4H**), could not develop such patterns well.

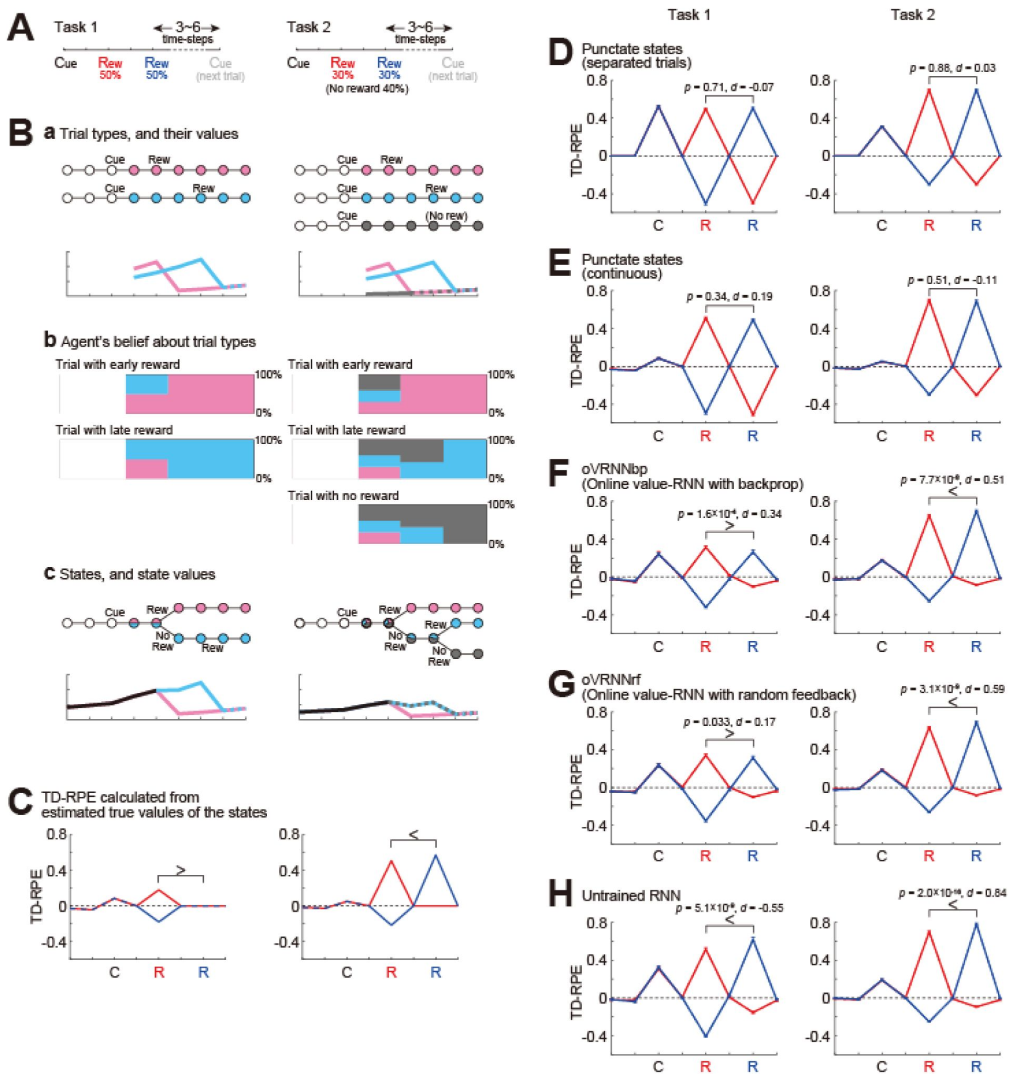

Figure 4

Simulation of two tasks having probabilistic structures, which were qualitatively similar to the two tasks examined in experiments [54] and modeled by the original value-RNN with BPTT [26].

(A) Simulated two tasks, in which reward was given at the early or the late timing with equal probabilities in all the trials (task 1) or 60% of trials (task 2). **(B) (a)** *Top*: Trial types. Two trial types (with early reward and with late reward) in task 1 and three trial types (with early reward, with late reward, and without reward) in task 2. *Bottom*: Value of each timing in each trial type estimated through simulations. **(b)** Agent's probabilistic belief about the current trial type, in the case where agent was in fact in the trial with early reward (top row), the trial with late reward (second row), or the trial without reward (third row in task 2). **(c)** *Top*: States defined by considering the probabilistic beliefs at each timing from cue. *Bottom*: True state/timing values calculated by taking (mathematical) expected value of the estimated value of each timing in each trial type. **(C)** Expected TD-RPE calculated from the estimated true values of the states/timings for task 1 (left) and task 2 (right). Red lines: case where reward was given at the early timing, blue lines: case where reward was given at the late timing. It is expected that TD-RPE at early reward is larger than TD-RPE at late reward in task 1 whereas the opposite is the case in task 2, as indicated by the inequality sings. **(D-H)** TD-RPEs at the latest trial within 1000 trials in which reward was given at the early timing (red lines) or the late timing (blue lines), averaged across 100 simulations (error-bars indicating ± SEM across simulations), in the different types of agent: (D,E) TD-RL agent having punctate state representation and state values without (D) or with (E) continuation between trials; (F) oVRNNbp. The number of RNN units was 12 (same applied to (G,H)); (G) oVRNNrf; (H) agent with untrained RNN. The p values are for paired t-test between TD-RPE at early reward and TD-RPE at late reward (100 pairs, two-tailed), and the d values are Cohen's d using an average variance and their signs are with respect to the expected patterns shown in (C) (same applied to **Fig. 8** and **Fig. 9C**).

3.1 Online value-RNN with further biological constraints

So far, the activities of neurons in the RNN ($\boldsymbol{x}$) were initialized to pseudo standard normal random numbers, and thereafter took numbers in the range between −0.5 and 0.5 that was the range of the sigmoidal input-output function. The value weights ($\boldsymbol{w}$) could also take both positive and negative values since no constraint was imposed. The fixed random feedback in oVRNNrf ($\boldsymbol{c}$) was generated by pseudo standard normal random numbers, and so could also be positive or negative. Negativity of the neurons' activities and the value weights could potentially be regarded as inhibitory or smaller-than-baseline quantities. However, because neuronal firing rate is non-negative and cortico-striatal projections are excitatory, it would be biologically more plausible to assume that the activities of neurons in the RNN and the value weights are non-negative. As for the fixed random feedback, if it is negative, the update rule becomes anti-Hebbian under positive TD-RPE, and so assuming non-negativity would be plausible since Hebbian property has been suggested for rapid plasticity of cortical synapses [55]. Regarding the connection weights in/onto the RNN, here we keep the original assumption that they could be positive or negative because it could be an approximate description of recurrent neuronal network with both recurrent excitation and inhibition. Later (Section 4.1) we will examine extended models that incorporate excitatory and inhibitory units and conform to the Dale's law.

Other than the sign of connection weights, there was another biological plausibility issue in the update rule for recurrent and feed-forward connections that were derived from the gradient descent. Specifically, the dependence on the post-synaptic activity was non-monotonic, maximized at the middle of the range of activity. It would be more biologically plausible to assume a monotonic increase (while an *opposite* shape of non-monotonicity, once decrease and thereafter increase, called the BCM (Bienenstock-Cooper-Munro) rule has actually been suggested [56–58]).

In order to address these issues, we considered revised models. We first considered a revised oVRNNbp (with backprop-type transported weights), referred to as oVRNNbp-rev, in which the RNN activities and the value weights were constrained to be non-negative, while the non-monotonic dependence of the update rule on the post-synaptic activity remained unchanged (**Fig. 5A**). We then considered a revised oVRNNrf, referred to as oVRNNrf-bio, in which the fixed random feedback, as well as the RNN activities and the value weights, were constrained to be non-negative, and also the update rule was modified so that the dependence on the post-synaptic activity became monotonic (with saturation) (**Fig. 5B**).

We examined how these revised models, in comparison with agent with untrained RNN that also had non-negative constraint for $\boldsymbol{x}$ and $\boldsymbol{w}$, performed in the Pavlovian cue-reward association task examined above (the numbers of RNN-units and trials were set to 12 and 1500, respectively). oVRNNbp-rev well developed state values toward reward (**Fig. 6A**). oVRNNrf-bio also developed state values to a largely comparable extent (**Fig. 6B**). By contrast, agent with untrained RNN could not develop such a pattern of state values (**Fig. 6C**). This, however, could be because initially set recurrent/feed-forward connections were far from those learned in the online value-RNNs. Therefore, as a more strict control, we conducted simulations of agent with untrained RNN with non-negative $\boldsymbol{x}$ and $\boldsymbol{w}$, where in each simulation the recurrent/feed-forward connections were set to be those shuffled from the learnt connections in a simulation of oVRNNrf-bio (hereafter we refer to these two types of untrained RNN as "naive untrained RNN" and "shuffled untrained RNN"). The model with shuffled untrained RNN developed state values somewhat better than the naive untrained RNN case (**Fig. 6D**), but still worse than oVRNNbp-rev and oVRNNrf-bio.

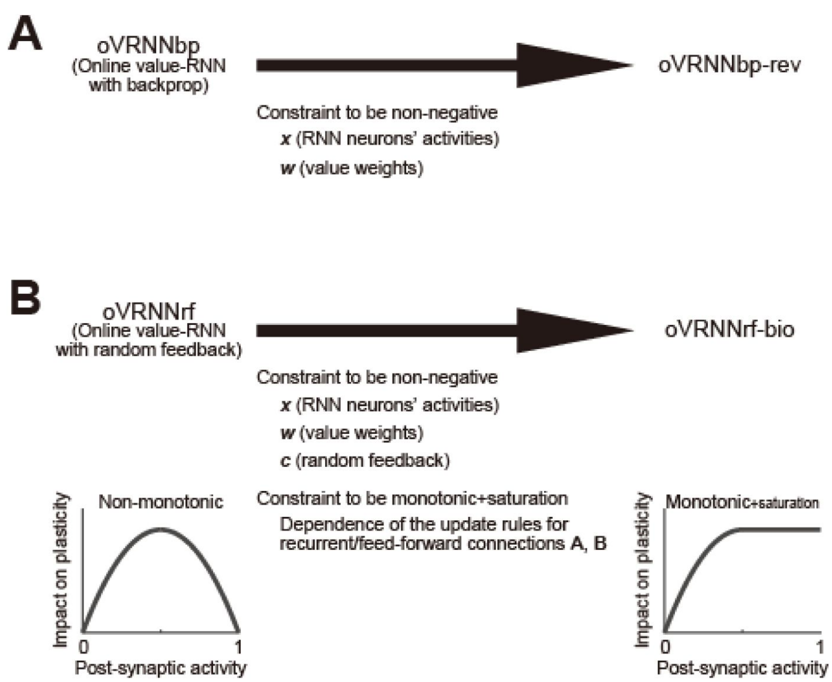

Figure 5

Revised online value-RNN models with further biological constraints.

(A) oVRNNbp-rev: oVRNNbp (online value-RNN with backprop) was modified so that the activities of neurons in the RNN (***x***) and the value weights (***w***) became non-negative. **(B)** oVRNNrf-bio: oVRNNrf (online value-RNN with fixed random feedback) was modified so that ***x*** and ***w***, as well as the fixed random feedback (***c***), became non-negative and also the dependence of the update rules for recurrent/feed-forward connections (**A** and **B**) on post-synaptic activity became monotonic + saturation.

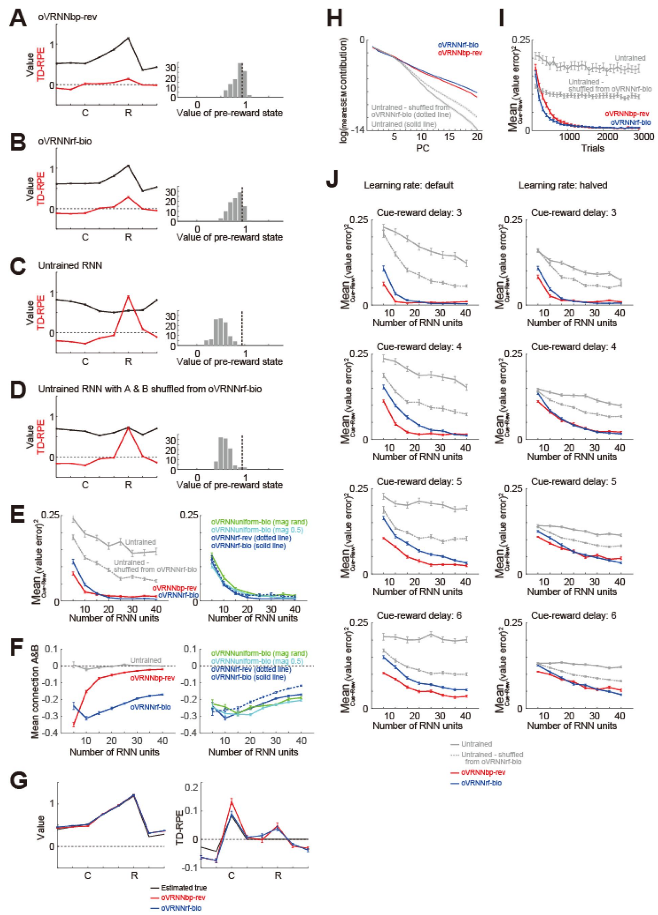

Figure 6

Performances of the revised online value-RNN models in the cue-reward association task, in comparison with models with untrained RNN that also had the non-negative constraint.

(A-D) State values (black lines) and TD-RPEs (red lines) at 1500-th trial in oVRNNbp-rev (A), oVRNNrf-bio (B), agent with naive untrained RNN (i.e., randomly initialized RNN) with ***x*** and ***w*** constrained to be non-negative (C), and agent with untrained RNN with connections shuffled from those learnt in oVRNNrf-bio and also with non-negative ***x*** and ***w*** (D). The number of RNN units was 12 in all the cases. Error-bars indicate mean ± SEM across 100 simulations; same applied to the followings unless otherwise mentioned. The right histograms show the across-simulation distribution of the value of the pre-reward state in each model. The vertical black dashed lines in the histograms indicate the true value of the pre-reward state (estimated through simulations). **(E)** *Left*: Mean squared value-error at 1500-th trial in oVRNNbp-rev (red line), oVRNNrf-bio (blue line), agent with naive untrained RNN (gray solid line: partly out of view), and agent with shuffled untrained RNN (gray dotted line) when the number of RNN units (n) was varied from 5 to 40. Learning rate for value weights was normalized by dividing by n/12 (same applied to the followings). *Right*: Mean squared value-error in oVRNNrf-bio (blue line: same data as in the left panel), oVRNN-bio with random-magnitude uniform feedback (green line), oVRNN-bio with fixed-magnitude (0.5) uniform feedback (light blue line), and oVRNNrf-rev where the update rule of oVRNNrf-bio was changed back to the original one (blue dotted line). **(F)** *Left*: Mean of the elements of the recurrent and feed-forward connections (at 1500-th trial) of oVRNNbp-rev (red line), oVRNNrf-bio (blue line), and naive untrained RNN (gray solid line). *Right*: Mean of the elements of the recurrent and feed-forward connections of oVRNNrf-bio (blue line: same data as in the left panel), oVRNN-bio with random-magnitude uniform feedback (green line), oVRNN-bio with fixed-magnitude (0.5) uniform feedback (light blue line), and oVRNNrf-rev (blue dotted line). **(G)** Learned state values (left panel) and TD-RPEs (right panel) in oVRNNbp-rev (red lines) and oVRNNrf-bio (blue lines) in the cases with 40 RNN units, compared to the estimated true values (black lines). **(H)** Log of contribution ratios of the principal components of the time series (for 1500 trials) of RNN activities in each model with 20 RNN units. **(I)** Mean squared value-error in each model with 20 RNN units across trials. **(J)** Mean squared value-error at 3000-th trial in each model in the cases where the cue-reward delay was 3, 4, 5, or 6 time-steps (top to bottom panels). Left and right panels show the results with default learning rates and halved learning rates, respectively.

3.2 Systematic simulations and analyses

We varied the number of RNN units (n), with the learning rate for value weights normalized by dividing by n/12, and compared the performance (mean of squared errors of state values between cue and reward at 1500-th trial) of oVRNNbp-rev and oVRNNrf-bio, in comparison with models with naive or shuffled untrained RNN. As shown in the left panel of **Fig. 6E**, oVRNNbp-rev and oVRNNrf-bio exhibited largely comparable performance and always outperformed the models with untrained RNN ($p < 2.5 \times 10^{-12}$ in Wilcoxon rank sum test for oVRNNbp-rev or oVRNNrf-bio vs naive or shuffled untrained for each number of RNN units), although oVRNNbp-rev somewhat outperformed or underperformed oVRNNrf-bio when the number of RNN units was small (≤10 ($p < 0.00029$)) or large (≥25 ($p < 3.7 \times 10^{-6}$)), respectively (**Figure 6G** shows the learned state values and TD-RPEs in oVRNNbp-rev and oVRNNrf-bio in the cases with 40 RNN units, compared to the estimated true values). Remarkably, oVRNNrf-bio generally achieved better performance than both oVRNNbp and oVRNNrf, which did not have the non-negative constraint (Wilcoxon rank sum test, vs oVRNNbp : $p < 7.8 \times 10^{-6}$ for 5 or ≥25 RNN units; vs oVRNNrf: $p < 0.021$ for ≤10 or ≥20 RNN units).

The left panel of **Figure 6F** shows the mean of the elements of the recurrent and feed-forward connections at 1500-th trial in the different models. As shown in this figure, these connections (initialized to pseudo standard normal random numbers) were learnt to become negative on average, in oVRNNbp-rev and oVRNNrf-bio. This learnt negative-dominance (inhibition-dominance) could possibly be related, e.g., through prevention of excessive activity, to the good performance of oVRNNrf-bio and also the better performance of the shuffled untrained RNN than the naive untrained RNN. The green and light blue lines in the right panels of **Figure 6E** and **Figure 6F** show the results for special cases where the random feedback in oVRNNrf-bio was fixed to the direction of $(1, 1, \ldots, 1)^{\mathrm{T}}$ (i.e., uniform feedback) with a random non-negative magnitude (green line) or a fixed magnitude of 0.5 (light blue line). The performance of these special cases, especially the former (with random magnitude) was somewhat worse than that of oVRNNrf-bio, but still better than that of the models with untrained RNN. The blue dotted lines in the right panels of **Figure 6E** and **Figure 6F** show the results where the modified update rule of oVRNNrf-bio was changed back to the original rule with non-monotonic dependence on the post-synaptic activity (**Fig. 5A**). The performance of this model was somewhat worse than oVRNNrf-bio, indicating that the biologically motivated modification of the update rule in fact improved the performance.

Figure 6H shows contribution ratios of PCs of the time series of RNN activities in each model with 20 RNN units. Compared with the cases with naive/shuffled untrained RNN, in oVRNNbp-rev and oVRNNrf-bio, later components had relatively high contributions (PC5~20 $p < 1.4 \times 10^{-6}$ (t-test vs naive) or < 0.014 (vs shuffled) in oVRNNbp-rev; PC6~20 $p < 2.0 \times 10^{-7}$ (vs naive) or PC7~20 $p < 5.9 \times 10^{-14}$ (vs shuffled) in oVRNNrf-bio), explaining their superior value-learning performance. **Figure 6I** shows how learning proceeded across trials in the models with 20 RNN units. While oVRNNbp-rev and oVRNNrf-bio eventually reached a comparable level of errors, oVRNNrf-bio outperformed oVRNNbp-rev in early trials (at 200, 300, 400, or 500 trials; $p < 0.049$ in Wilcoxon rank sum test for each). This is presumably because the value weights did not develop well in early trials and so the backprop-type feedback, which was the same as the value weights, did not work well, while the non-negative fixed random feedback worked finely from the beginning. **Figure 6J** shows the cases with longer cue-reward delays, with default or halved learning rates. As the delay increased, the mean squared error of state values (at 3000-th trial) increased, but the relative superiority of oVRNNbp-rev and oVRNNrf-bio over the models with untrained RNN remained to hold, except for a few cases with 5 RNN units (5 delay oVRNNrf-bio vs shuffled with default learning rate, 6 delay oVRNNrf-bio vs naive or shuffled with halved learning rate) ($p < 0.047$ in Wilcoxon rank sum test for oVRNNbp-rev or oVRNNrf-bio vs naive or shuffled untrained for each number of RNN units for each delay).

We further examined how the revised online value-RNN models performed in the two tasks with probabilistic structures examined above. The models with 12 RNN units appeared not able to produce the expected different patterns of TD-RPEs in the two tasks (TD-RPE at early reward > TD-RPE at late reward in task 1 and opposite pattern in task 2), and we increased the number of RNN units to 20. Then, both oVRNNbp-rev and oVRNNrf-bio produced such TD-RPE patterns (**Fig. 7A,B**) whereas the models with untrained RNN could not (**Fig. 7C,D**). This indicates that the online value-RNN with random feedback and further biological constraints could learn the differential characteristics of the tasks.

3.3 Loose alignment and feedback alignment

Coming back to the original cue-reward association task, we examined how the angle between the value weights ($\boldsymbol{w}$) and the random feedback ($\boldsymbol{c}$) changed across trials in oVRNNrf-bio with 12 RNN units. As shown in **Fig. 8A**, the angle was on average smaller than 90°, which was the chance-level angle in the case without non-negative constraint, from the beginning, while there was no further alignment over trials. This could be understood as follows. Because both the value weights ($\boldsymbol{w}$) and the random feedback ($\boldsymbol{c}$) were now constrained to be non-negative, these two vectors were ensured to be in a relatively close angle (i.e., in the same quadrant) from the beginning. By virtue of this loose alignment, the random feedback could act similarly to backprop-type transported-weight feedback, even without further alignment.

We examined if the angle between the value weights ($\boldsymbol{w}$) and the random feedback ($\boldsymbol{c}$) at 1500-th trial was associated with the developed value of pre-reward state across simulations, but found no association (r = 0.0117, p = 0.908) (**Fig. 8B**). We then examined if the $\boldsymbol{w}$-$\boldsymbol{c}$ angle at earlier trials (2^{nd} - 500-th trials) was associated with the developed values at 500-th trial, with the number of simulations increased to 1000 so that small correlation could be detected. We found that the $\boldsymbol{w}$-$\boldsymbol{c}$ angle at initial trials (2nd - around 10-th trials) was negatively correlated with the developed values of the reward state and preceding states at 500-th trial (**Fig. 8C**). As for the reward state, negative correlation at around 100-th - 300-th trial was also observed. These results suggest that better alignment of $\boldsymbol{w}$ and $\boldsymbol{c}$ at initial and early timings was associated with better development of state values, in line with the conjecture that loose alignment of $\boldsymbol{w}$ and $\boldsymbol{c}$ coming from the non-negative constraint supported learning. It should be noted, however, that there were cases where positive (although small) correlation was observed. Its exact reason is not sure, but it could be related to the fact that largeness of developed values or the speed of value development does not necessarily mean good learning.

As mentioned above, while the angle between $\boldsymbol{w}$ and $\boldsymbol{c}$ was on average smaller than 90° from the beginning, there was no further alignment over trials. This seemed mysterious because the mechanism for feedback alignment that we derived for the models without non-negative constraint was expected to work also for the models with non-negative constraint. As a possible reason for the non-occurrence of feedback alignment, we guessed that one or a few element(s) of $\boldsymbol{w}$ grew prominently during learning, and so $\boldsymbol{w}$ became close to an edge or boundary of the non-negative quadrant and thereby angle between $\boldsymbol{w}$ and other vector became generally large (as illustrated in **Fig. 8D**). **Figure 8Ea** shows the mean±SEM of the elements of $\boldsymbol{w}$ ordered from the largest to smallest ones after 1500 trials. As conjectured above, a few elements indeed grew prominently.

We considered that if a slight decay (forgetting) of value weights (c.f., [59]–[61]) was assumed, such a prominent growth of a few elements of $\boldsymbol{w}$ may be mitigated and alignment of $\boldsymbol{w}$ to $\boldsymbol{c}$, beyond the initial loose alignment because of the non-negative constraint, may occur. These conjectures were indeed confirmed by simulations (**Fig. 8Eb,c** and **Fig. 8F**). The mean squared value error slightly increased when the value-weight-decay was assumed (**Fig. 8G**), however, presumably reflecting a decrease in developed values and a deterioration of learning because of the decay.

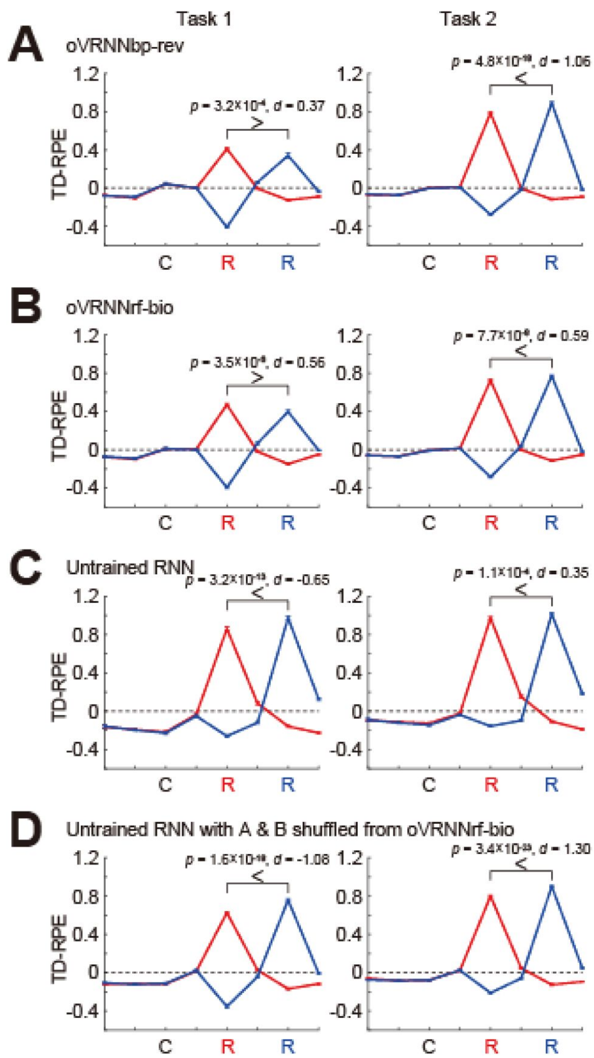

Figure 7

Performances of the revised online value-RNN models with further biological constraints in the two tasks having probabilistic structures, in comparison with models with untrained RNN.

TD-RPEs at the latest trial within 2000 trials in which reward was given at the early timing (red lines) or the late timing (blue lines) in task 1 (left) and task 2 (right), averaged across 100 simulations (error-bars indicating ± SEM across simulations), are shown for the four types of agent: (A) oVRNNbp-rev; (B) oVRNNrf-bio; (C) agent with naive untrained RNN; (D) agent with untrained RNN with connections shuffled from those learnt in oVRNNrf-bio. The number of RNN units was 20 for all the cases.

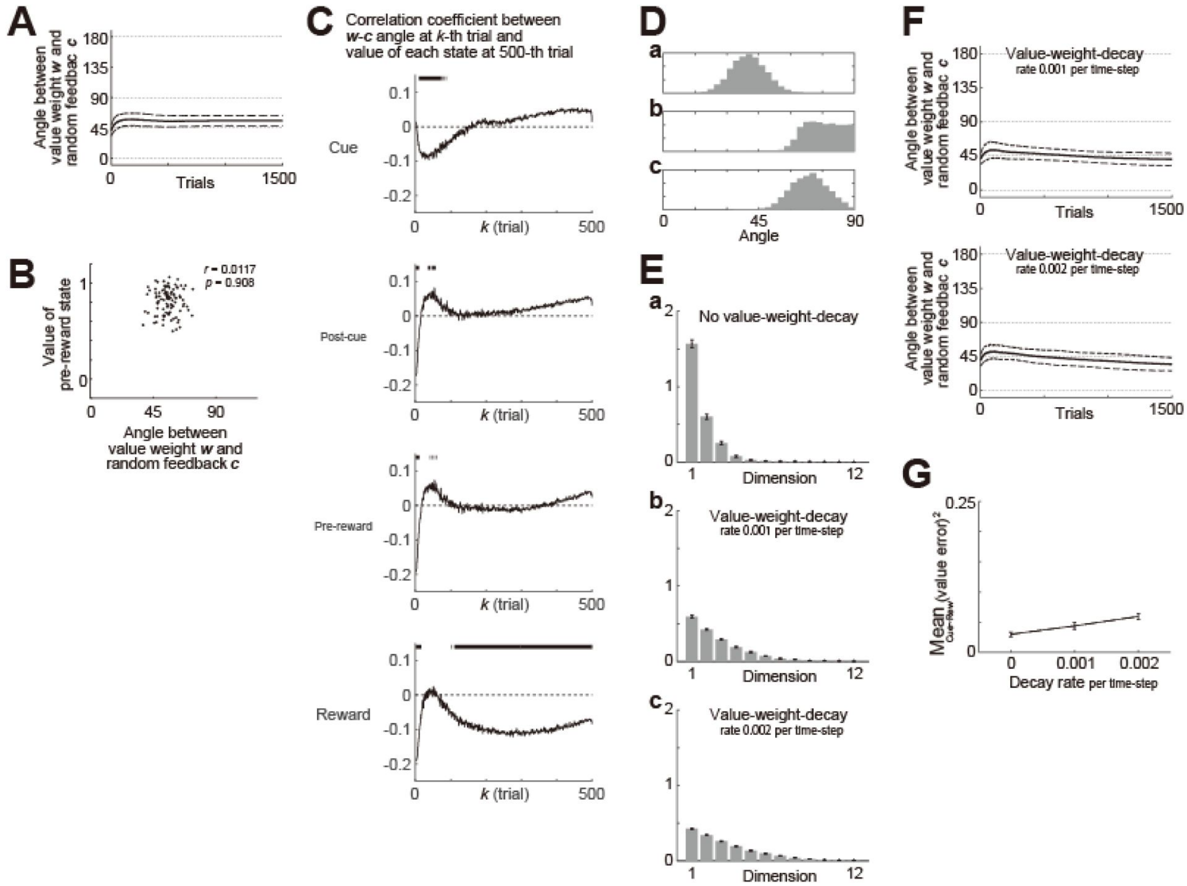

Figure 8

Loose alignment of the value weights (*w*) and the random feedback (*c*) in oVRNNrf-bio (with 12 RNN units).

(A) Over-trial changes in the angle between the value weights ***w*** and the fixed random feedback ***c***. The solid line and the dashed lines indicate the mean and ± SD across 100 simulations, respectively. **(B)** No correlation between the ***w***-***c*** angle (horizontal axis) and the value of the pre-reward state (vertical axis) at 1500-th trial (r = 0.0117, p = 0.908). The dots indicate the results of individual simulations. **(C)** Correlation between the ***w***-***c*** angle at k-th trial (horizontal axis) and the value of the cue, post-cue, pre-reward, or reward state (top-bottom panels) at 500-th trial across 1000 simulations. The solid lines indicate the correlation coefficient, and the short vertical bars at the top of each panel indicates the cases in which p-value was less than 0.05. **(D)** Distribution of the angle between two 12-dimensional vectors when the elements of both vectors were drawn from [0 1] uniform pseudo-random numbers (a) or when one of the vectors was replaced with [1 0 0 … 0] (i.e., on the edge of the non-negative quadrant) (b) or [1 1 0 … 0] (i.e., on the boundary of the non-negative quadrant) (c). **(E)** Across-simulations histograms of elements of ***w*** in oVRNNrf-bio with 12 RNN units ordered from the largest to smallest ones after 1500 trials when there was no value-weight-decay (a) or there was value-weight-decay with decay rate (per time-step) of 0.001 (b) or 0.002 (c). The error-bars indicate the mean ± SEM across 100 simulations. **(F)** Over-trial changes in the angle between the value weights ***w*** and the fixed random feedback ***c*** when there was value-weight-decay with decay rate (per time-step) of 0.001 (top panel) or 0.002 (bottom panel). Notations are the same those in (A). **(G)** Mean squared value-error at 1500-th trial in oVRNNrf-bio with 12 RNN units with the rate of value-weight-decay was varied (horizontal axis). The error-bars indicate the mean ± SEM across 100 simulations.

4.1 Models with excitatory and inhibitory units

As mentioned above, in oVRNNbp-rev and oVRNNrf-bio, the connection weights in/onto the RNN could be both positive and negative, against the Dale's law. Recent studies started to examine neural networks incorporating the Dale's law [62], [63] or other connectivity features [64]. So we examined extended models, named oVRNNbp-rev-ei and oVRNNrf-bio-ei, which incorporated excitatory E-units, modeling pyramidal cells, and inhibitory I-units, modeling fast-spiking (FS) cells (**Fig. 9A**). Cortical excitation can operate slowly due to slow synaptic dynamics [46], [47] (see the description about the time-step in the Methods for details). In contrast, inhibition from FS cells to pyramidal cells may operate more quickly, since it was shown [65] that observed phases of regular-spiking (RS, putatively pyramidal) cells' and FS cells' spikes [66] could be explained by fast FS→RS inhibition and temporally distributed recurrent excitation.

Given these, we assumed that excitation from E-units to E- and I-units took one time-step whereas I→E inhibition operated within a time-step, and also that each E-unit received inputs from all the E-units and a particular I-unit (although this assumption could be supported by the abovementioned suggestions, its validity remains largely open). Chemical and electrical connections between FS cells exist and are suggested to serve for synchronization or oscillation [67], [68], but we omitted I→I connections because our models did not describe fast spike dynamics. Since FS cells can fire at high frequencies, we assumed that the activation function for I-units was not saturating but linear. Lastly, we did not assume plasticity for connections from/to I-units. The connection weights onto the E- and I-units from the observation units and E-units were non-negatively initialized, specifically, initialized to pseudo normal random numbers with mean = 3 and standard deviation = 1 and rectified to 0 when becoming negative.

We examined how these extended models behaved in the Pavlovian task and the probabilistic tasks. As shown in **Fig. 9B**, oVRNNbp-rev-ei and oVRNNrf-bio-ei learned the state values in the Pavlovian task much more accurately than the models with naive untrained RNN or untrained RNN whose connections from the observation and E-units to E- and I-units were shuffled from oVRNNrf-bio-ei ($p < 5.2\times10^{-12}$ in Wilcoxon rank sum test for oVRNNbp-rev-ei or oVRNNrf-bio-ei vs naive or shuffled untrained for each number of RNN units). oVRNNbp-rev-ei somewhat outperformed or underperformed oVRNNrf-bio-ei when the number of RNN units was small (≤10 ($p < 0.0091$)) or relatively large (15~35 ($p < 0.027$)), respectively. Also, as shown in **Fig. 9C**, oVRNNbp-rev-ei and oVRNNrf-bio-ei with 20 E-units and 20 I-units generated the expected different patterns of TD-RPEs in the two tasks (TD-RPE at early reward > TD-RPE at late reward in task 1 and opposite pattern in task 2) although the effect size for task 2 in oVRNNrf-bio-ei was small, while the models with untrained RNN did not.

As such, the extended models with E-units and I-units showed largely similar behaviors to those of the original oVRNNbp-rev and oVRNNrf-bio with mixed positive and negative RNN weights. This is actually reasonable, because combining the update equations for I-units and E-units in the extended models (top two equations in **Fig. 9A**) results in an equation largely similar to the update equation for RNN units in the original models (top equation in **Fig. 1**). In other words, the original models with mixed positive and negative RNN weights could be regarded as a simplified description of the models with E-units and I-units under the abovementioned assumptions. Therefore, for simplicity, we will return to mixed positive and negative RNN weights in the following.

4.2 Task with distractor cue

So far we have examined situations where there existed a reward and a cue associated with the reward. However, in real environments, it is likely that there exist both reward-associated and non-associated (distractor) cues and agent does not initially know which cue is associated with reward and which is not. Learning cue-reward association in such distractor-existing

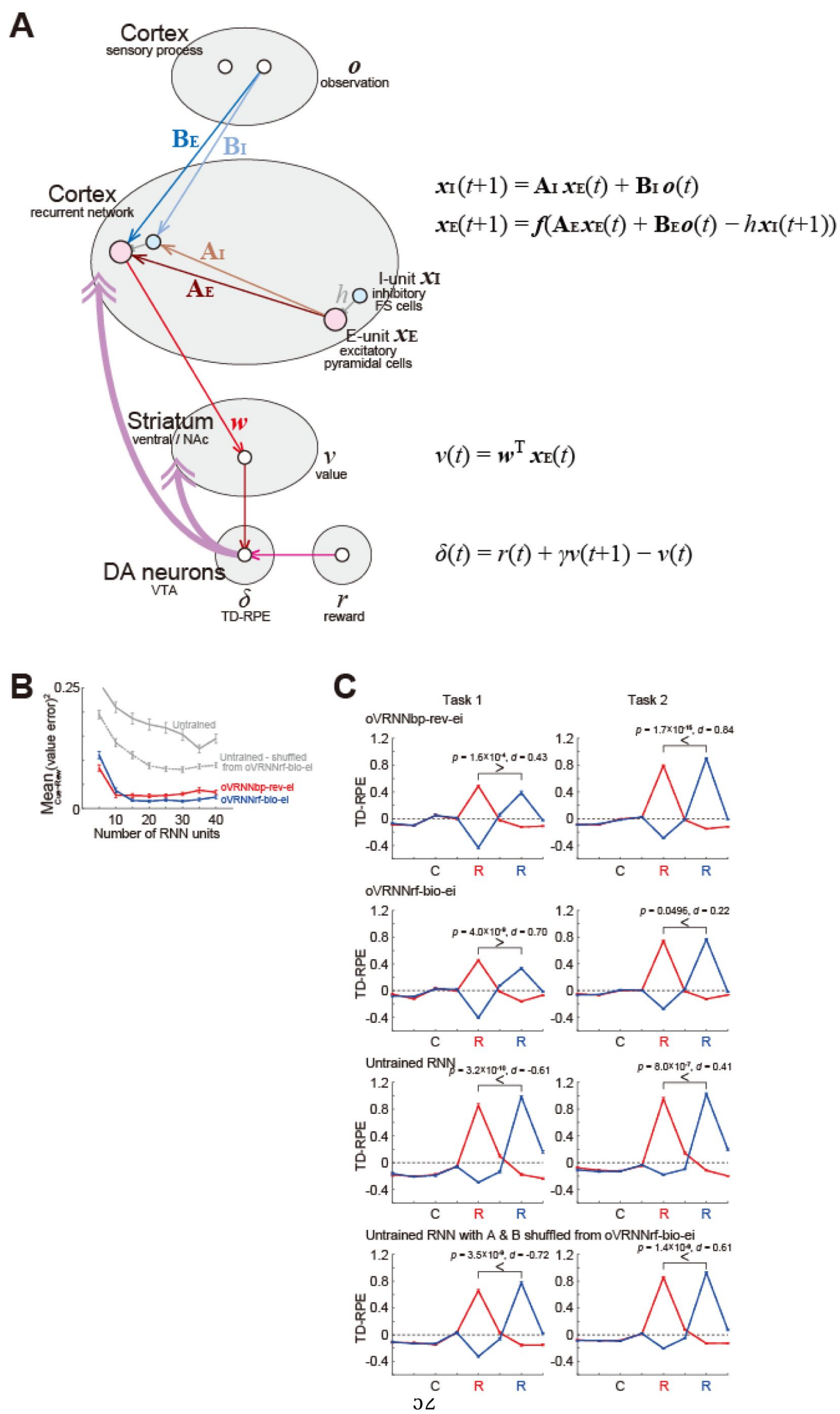

Figure 9

oVRNNbp-rev-ei and oVRNNrf-bio-ei models incorporating excitatory E-units and inhibitory I-units.

(A) Schematic illustration of the models' architecture. For ease of viewing, only limited parts of units and connections are drawn. **(B)** Mean squared value-error at 1500-th trial in the cue-reward association task in oVRNNbp-rev-ei (red line), oVRNNrf-bio-ei (blue line), and E-/I-units-incorporated models with naive untrained RNN (i.e., randomly initialized RNN) (gray solid line) or untrained RNN with connections shuffled from those learnt in oVRNNrf-bio-ei (gray dotted line). **(C)** Patterns of TD-RPE in the tasks with probabilistic structures generated in the four models with E-/I-units. Simulation conditions and notations are the same those in **Fig. 7**.

environments is generally not easy for biologically constrained models, and it has been addressed by only few previous work [69]. We examined whether our biologically constrained oVRNNrf-bio, as well as oVRNNbp-rev, could learn the cue-reward association under the presence of distractor cue. We considered a simple case where there existed a distractor cue, which was presented to the agent with a certain probability at every time-step, between cue-reward duration or reward-cue duration (i.e., inter-trial interval) or simultaneously with cue or reward. As for the agent's models, we assumed that the observation inputs had an additional element (dimension), which was set to 1 when the distractor was presented and 0 when not (**Fig. 10A**).

We examined how oVRNNbp-rev, oVRNNrf-bio, and the models with untrained RNN behaved in the modified Pavlovian task with a distractor cue, which was presented with probability 0, 0.1, 0.2, or 0.3 at every time-step (**Fig. 10B-E**, left panels). As a result, even when there was such a distractor cue, oVRNNbp-rev and oVRNNrf-bio could still learn the state values better than the models with naive or shuffled untrained RNN ($p < 1.7\times10^{-10}$ in Wilcoxon rank sum test for oVRNNbp-rev or VRNNrf-bio vs naive or shuffled untrained for each number of RNN units for each level of distractor probability) (**Fig. 10C-E**, middle panels), although the accuracy moderately decreased compared with the case without distractor cue (**Fig. 10B**, middle panel). These results suggest robustness of the learning ability of oVRNNrf-bio against distractor in realistic situations. We further examined how the models with E- and I-units behaved in the task with a distractor cue, and confirmed that even in the presence of disctractor cue, oVRNNbp-rev-ei and oVRNNrf-bio-ei could learn the state values better than the models with naive or shuffled untrained RNN ($p < 3.2\times10^{-8}$ in Wilcoxon rank sum test for oVRNNbp-rev-ei or oVRNNrf-bio-ei vs naive or shuffled untrained for each number of RNN units for each level of distractor probability) (**Fig. 10B-E**, right panels).

4.3 Incorporation of action selection

Ultimate purpose of animals and RL agents is to optimize their policy, i.e., probability of action selection at each state to maximize rewards. Therefore, we examined if our models could be extended to incorporate action selection. In reference to the proposals that algorithm akin to the actor-critic method may be implemented in the brain [35, 70, 71], we considered extended models oVRNNbp-rev-ac and oVRNNrf-bio-ac, which incorporated an actor-critic architecture (**Fig. 11A**). Specifically, each RNN unit was assumed to connect to not only the state-value(v)-representing unit in the ventral striatum but also the action-value(q_k)-representing units in the dorsal striatum. Their non-negative weights (u_{kj}) represented (as a vector) action preferences, which slightly decayed with time so as not to unboundedly increase.

It has been suggested that action is selected through competition of neural populations in the striatum-thalamus-cortex(-striatum) circuit, which represent or receive action values, in the presence of noise [72–74]. However, because our models did not describe fast neural dynamics (see the description about the time-step in the Methods), we assumed a soft-max function for action selection, i.e., assumed that an action was selected according to a soft-max probability determined by the difference in the action values when there were two action candidates. We then assumed that there is a cortical region, which contains action-representing neural populations, implemented as 'action units' in the model (**Fig. 11A**). Action unit was assumed to become active (i.e., = 1) when the corresponding action was selected and inactive (i.e., = 0) otherwise, and these action units were assumed to send inputs to the RNN, similarly to the observation units informing the presence of cue and reward.

We examined how these models (oVRNNbp-rev-ac and oVRNNrf-bio-ac) and control models with naive or shuffled untrained RNN behaved in two-alternative choice tasks. In the first choice task (**Fig. 11Ba**), taking action 1 lead to a large (size 2) reward two time-steps later whereas taking action 2 lead to a small (size 1) reward two time-steps later. oVRNNbp-rev-ac and oVRNNrf-bio-ac, after training, successfully selected the better action, action 1, in most cases, whereas the models

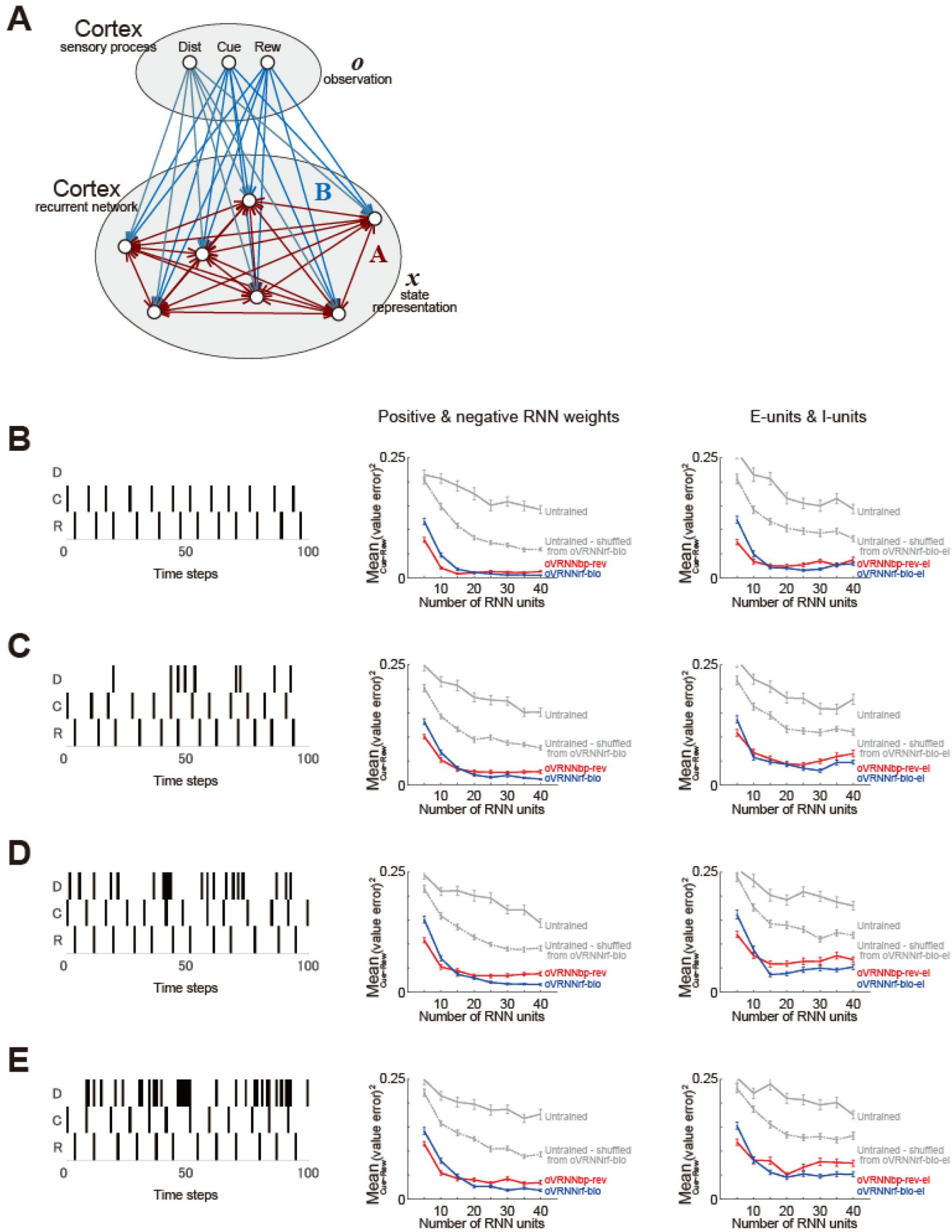

Figure 10

Cue-reward association task with distractor cue.

(A) Modification of oVRNNbp-rev and oVRNNrf-bio to incorporate possible existence of distractor cue. The observation units ***o*** had an additional element (leftmost circle labeled as 'Dist'), which was 1 at the time-steps where distractor cue was present and 0 otherwise. **(B-E)** Results of the cases where the probability of the presence of distractor cue at every time-step was 0 (B), 0.1 (C), 0.2 (D), and 0.3 (E). *Left panels*: Examples of the presence of distractor ('D'), reward-associated cue ('C'), and reward ('R') over 100 time-steps. *Middle panels*: Mean squared value-error at 1500-th trial in oVRNNbp-rev (red line), oVRNNrf-bio (blue line), and the models with naive or shuffled untrained RNN (gray solid or dotted line). *Right panels*: Results for the models with E- and I-units (oVRNNbp-rev-ei: red line, oVRNNrf-bio-ei: blue line, models with naive or shuffled untrained RNN: gray solid or dotted line), which were modified to incorporate possible existence of distractor cue in the same manner as in (A).

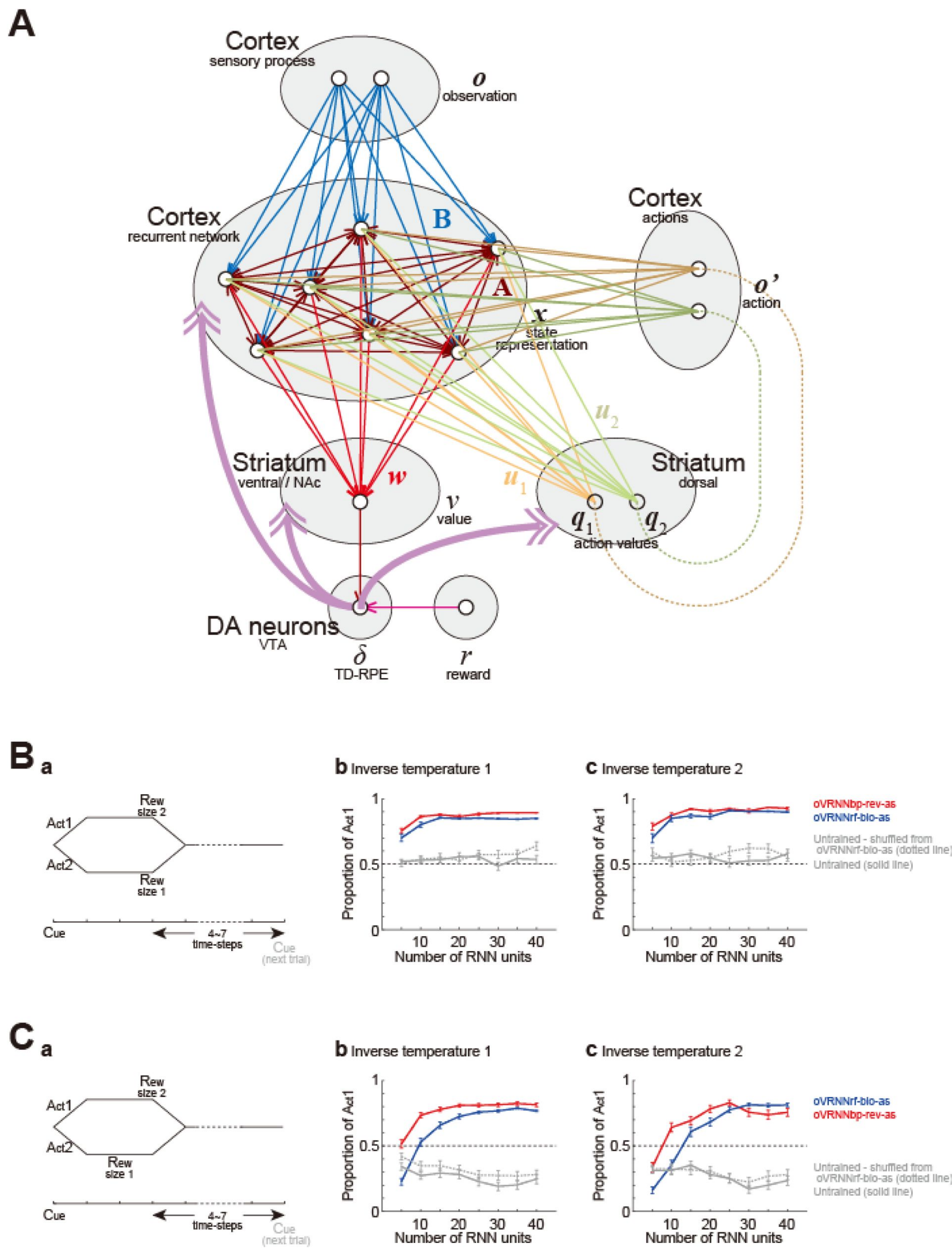

Figure 11

Incorporation of action selection. (A) Schematic illustration of the models incorporating an actor-critic architecture.

(B) Two-alternative choice task. **(a)** Task diagram. **(b,c)** Proportion of Action 1 selection in 2901∼3000-th trials in oVRNNbp-rev-as (red line), oVRNNrf-bio-as (blue line), and the models with naive untrained RNN (i.e., randomly initialized RNN) (gray solid line) or untrained RNN with connections shuffled from those learnt in VRNNrf-bio-as (gray dotted line). Error-bars indicate the mean ± SEM over 100 simulations. The inverse temperature was set to 1 (b) or 2 (c). **(C)** Inter-temporal choice task. The notations are the same as those in (B).

with untrained RNN did not develop strong tendency to select action 1 (**Fig. 11Bb,c**). Next, in the second choice task (**Fig. 11Ca**), taking action 1 lead to a large (size 2) reward two time-steps later whereas taking action 2 lead to a small (size 1) reward one time-step later. So this task imposed inter-temporal choice between delayed large reward and sooner small reward. oVRNNbp-rev-ac, after training, tended to select action 1, which was better than action 2 even when presumed temporal discounting (0.8 per time-step) was taken into account (**Fig. 11Cb,c**). oVRNNrf-bio-ac also tended to select action 1 when the number of RNN units was not small. On the other hand, the models with untrained RNN tended to select sooner small reward. These results suggest that oVRNNrf-bio-ac, as well as oVRNNbp-rev-ac, could learn to select advantageous action to a certain extent, even in the case of inter-temporal choice.

Discussion

We have shown that state representation and value can be learned online in the RNN and its readout by using random feedback instead of biologically unavailable downstream weights. This was achieved through feedback alignment, and we have presented an intuitive understanding of its mechanism. We have further shown that the non-negative constraint realizes loose alignment of the forward weights and feedback from the beginning, which appeared to support learning.

Roles of DA

Midbrain DA neurons project to both striatum and cortex, including the prefrontal cortex [75] and the hippocampus [76]. As for striatal DA, DA-dependent cortico-striatal plasticity is considered to implement TD-RPE-based value update [7, 77]. By contrast, while cortical DA has been implicated in working memory [78–81], decision making [82], and aversive memory [83], its role as TD-RPE remains unclear, despite findings suggesting that cortical DA does encode (TD-)RPE [84–87] and modulate plasticity [88, 89]. Learnability of our biologically constrained online value-RNN suggests that TD-RPE-encoding cortical DA modulates plasticity of RNN so that appropriate state representation can be learnt.

Many studies reported heterogeneities of DA signals, which may come from encoding prediction errors other than RPE or feature-specific components of RPE [90]. Referring to a result [91] indicating DA's encoding of non-reward PEs and the fact that DA neurons receive inputs from the cerebellum [92, 93], which presumably implements supervised learning [94], a recent work [45] proposed that DA encodes vector-valued errors used for supervised learning of actions in continuous space. In contrast, we assumed DA's encoding of scalar TD-RPE, which can be consistent with the heterogeneity due to encoding of feature-specific RPE components [90]. The previous model [45] and our model can coexist, with different DA neuronal populations encoding different types of errors, or a single DA neuron switching its encoding depending on context/inputs.

VTA DA neurons project also to the basolateral amygdala (BLA) [95], and DA regulates plasticity also there [96]. Moreover, VTA→BLA DA entailed properties of TD-RPE, although increased also upon aversive event, and was not itself reinforcing but crucial for the formation of environmental model [97]. BLA has recurrent connections [98], projects to the striatum [99, 100], and engages in abstract context representation [101]. Thus, given that goal-directed-like behavior could be achieved through sophisticated state representation [10–12], it could potentially be learned by value-RNN-like mechanism in the BLA. Whether such sophisticated representation can be learned, however, remains open, and it might require multidimensional error [102] beyond TD-RPE.

There are many DA-related mechanisms that were not incorporated into our models, including the distinctions of D1-direct and D2-indirect pathways [103–107] and cortical projections to them [108–111], as well as mechanisms underlying TD-RPE encoding (c.f., [106, 112]) or learning for it (c.f., [69]). Future studies are expected to incorporate these.

Predictions and implications of our models

oVRNNrf predicts that the feedback vector $\boldsymbol{c}$ and the value-weight vector $\boldsymbol{w}$ become gradually aligned, while oVRNNrf-bio predicts that $\boldsymbol{c}$ and $\boldsymbol{w}$ are loosely aligned from the beginning. Element of $\boldsymbol{c}$ could be measured as the magnitude of pyramidal cell's response to DA stimulation. Element of $\boldsymbol{w}$ corresponding to a given pyramidal cell could be measured, if striatal neuron that receives input from that pyramidal cell can be identified (although technically demanding), as the magnitude of response of the striatal neuron to activation of the pyramidal cell. Then, the abovementioned predictions could be tested by (i) identify cortical, striatal, and VTA regions that are connected, (ii) identify pairs of cortical pyramidal cells and striatal neurons that are connected, (iii) measure the responses of identified pyramidal cells to DA stimulation, as well as the responses of identified striatal neurons to activation of the connected pyramidal cells, and (iv) test whether DA→pyramidal responses and pyramidal→striatal responses are associated across pyramidal cells, and whether such associations develop through learning.

Testing this prediction, however, would be technically quite demanding, as mentioned above. An alternative way of testing our model is to manipulate the cortical DA feedback and see if it will cause (re-)alignment of value weights (i.e., cortical striatal strengths). Specifically, our model predicts that if DA projection to a particular cortical locus is silenced, effect of the activity of that locus on the value-encoding striatal activity will become diminished.

We have shown that oVRNNrf and oVRNNrf-bio could work even when the random feedback was uniform, i.e., fixed to the direction of $(1, 1, \ldots, 1)^{\mathrm{T}}$, although the performance was somewhat worse. This is reasonable because uniform feedback can still encode scalar TD-RPE that drives our models, in contrast to a previous study [45], which considered DA's encoding of vector error and thus regarded uniform feedback as a negative control. If oVRNNrf/oVRNNrf-bio-like mechanism indeed operates in the brain and the feedback is near uniform, alignment of the value weights $\boldsymbol{w}$ to near $(1, 1, \ldots, 1)$ is expected to occur. This means that states are (learned to be) represented in such a way that simple summation of cortical neuronal activity approximates value, thereby potentially explaining why value is often correlated with regional activation (fMRI BOLD signal) of cortical regions [113]. Notably, uniform feedback coupled with positive forward weights was shown to be effective also in supervised learning of one-dimensional output in feed-forward network [114], and we guess that loose alignment may underlie it.

On the RNN unit

In our oVRNNbp without non-negative constraint, as the number of RNN units increased, the squared error initially decreased but then increased (**Fig. 2J**) (while intriguingly it was not the case for the models with non-negative constraint (**Fig. 6E**)). In contrast, in the original value-RNN [12, 26], the ability to develop belief-state-like representation was reported to improve as the number of RNN units increased to 100 or 50. There are at least two possible reasons for this difference, other than the difference in the performance measures. First one is the difference in the update rules. As mentioned earlier, the original value-RNN used BPTT [27] whereas our oVRNNbp used an online learning rule, which only considered the influence of the recurrent weights at the previous time step.

Second one is a difference in the RNN unit. Specifically, the original value-RNN used the Gated Recurrent Unit (GRU) cell [115] whereas we used a simple sigmoidal function. RNN with simple nonlinear unit is known to have the vanishing gradient problem [116], which could be alleviated by using memorable/gated RNN such as the Long Short-Term Memory (LSTM) unit [117] or the GRU cell [115]. We used the simple sigmoidal unit because biological plausibility of the GRU cell appeared elusive. However, gated unit similar to the LSTM unit has actually been proposed to be implemented in cortical microcircuits [118], and incorporation of such biologically plausible gated unit into our online value-RNN would be a hopeful direction.

From a bottom-up viewpoint, our RNN unit did not incorporate spiking [41, 57, 119] nor nonlinear dendritic computations [44, 120, 121]. Recent studies suggest that dendritic mechanisms [37, 38], possibly in combination with burst-dependent plasticity [41, 42], can realize credit assignment without backprop in supervised and unsupervised learning [122, 123]. Also, recent model of hippocampus [25] has shown that a network of multi-compartment units could learn complex representations. Having dendritic mechanisms is different from just increasing the number of neural-network layers because of their own specific features/constraints, and it was argued [44] that adding such biological constraints enables learning in deep neural networks. Therefore, incorporation of biological details into RNN unit in our models would be hopeful also from the bottom-up viewpoint.

Comparison to other algorithms

As an alternative to backprop in hierarchical network, aside from feedback alignment [36], Associative Reward-Penalty ($A_{R\text{-}P}$) algorithm has been proposed [124–126]. In $A_{R\text{-}P}$, the hidden units behave stochastically, allowing the gradient to be estimated via stochastic sampling. Recent work [127] has proposed Phaseless Alignment Learning (PAL), in which high-frequency noise-induced learning of feedback projections proceeds simultaneously with learning of forward projections using the feedback in a lower frequency. Noise-induced learning of the weights on readout neurons from untrained RNN by reward-modulated Hebbian plasticity has also been demonstrated [128]. Such noise- or perturbation-based [40] mechanisms are biologically plausible because neurons and neural networks can exhibit noisy or chaotic behavior [129–131], and might improve the performance of value-RNN if implemented.

Regarding learning of RNN, "e-prop" [35] was proposed as a locally learnable online approximation of BPTT [27], which was used in the original value RNN [26]. In e-prop, neuron-specific learning signal is combined with weight-specific locally-updatable "eligibility trace". Reward-based e-prop was also shown to work [35], both in a setup not introducing TD-RPE with symmetric or random feedback (their Supplementary Figure 5) and in another setup introducing TD-RPE with symmetric feedback (their Figures 4 and 5). Compared to these, our models differ in multiple ways.

First, we have shown that alignment to random feedback occurs in the models driven by TD-RPE. Second, our models do not have "eligibility trace" (nor memorable/gated unit, different from the original value-RNN [26]), but could still solve temporal credit assignment to a certain extent because TD learning is by itself a solution for it (notably, recent work showed that combination of TD(0) and model-based RL well explained rat's choice and DA patterns [132]). However, as mentioned before, single time-step in our models was assumed to correspond to hundreds of milliseconds, incorporating slow synaptic dynamics, whereas e-prop is an algorithm for spiking neuron models with a much finer time scale. From this aspect, our models could be seen as a coarse-time-scale approximation of e-prop. On top of these, our results point to a potential computational benefit of biological non-negative constraint, which could effectively limit the parameter space and promote learning.

Methods

Online value-RNN with backprop (oVRNNbp)

We constructed an online value-RNN model based on the previous proposals [12], [26] but with several differences. We assumed that the activities of neurons in the RNN at time t+1 were determined by the activities of these neurons and neurons representing observation (cue, reward, or nothing) at time t:

$$\boldsymbol{x}(t+1) = \boldsymbol{f}(\mathbf{A}\boldsymbol{x}(t) + \mathbf{B}\boldsymbol{o}(t)), \quad \text{(Eq. 1.1)}$$

where

$\boldsymbol{x} = (x_j)$: activity of j-th neuron in the RNN ($j = 1, .., n$)
$\boldsymbol{o} = (o_k)$: activity of k-th neuron in the observation layer ($k = 1, 2$)
if there was a cue at t, $\boldsymbol{o}(t) = (1\ 0)^{\mathrm{T}}$,
if there was a reward at t, $\boldsymbol{o}(t) = (0\ 1)^{\mathrm{T}}$,
and otherwise, $\boldsymbol{o}(t) = (0\ 0)^{\mathrm{T}}$
$\mathbf{A} = (A_{ij})$: recurrent connection strength from x_j to x_i
$\mathbf{B} = (B_{ik})$: feed-forward connection strength from o_k to x_i
$f(z) = 1/(1 + \exp(-z)) - 0.5$: (Eq. 1.2)

sigmoidal function representing neuronal input-output relation

The estimated value of the state at t was calculated as

$$v(t) = \boldsymbol{w}^{\mathrm{T}}\, \boldsymbol{x}(t) \quad \text{(Eq. 1.3)}$$

where

$$\boldsymbol{w} = (w_j)$$

were the value weights. The error between this estimated value and the true value, $v_{true}(t)$, was defined as:

$$\varepsilon(t) = v_{true}(t) - v(t) \quad \text{(Eq. 1.4)}$$

Parameters w_j, A_{ij}, and B_{ik} that minimize the squared error $\varepsilon(t)^2$ could be found by a gradient descent / error-backpropagation (backprop) method, i.e., by updating them in the directions of $-\partial(\varepsilon(t)^2)/\partial w_j$, $-\partial(\varepsilon(t)^2)/\partial A_{ij}$, and $-\partial(\varepsilon(t)^2)/\partial B_{ik}$. $-\partial(\varepsilon(t)^2)/\partial w_j$ was calculated as follows:

$-\partial(\varepsilon(t)^2)/\partial A_{ij}$, and $-\partial(\varepsilon(t)^2)/\partial B_{ik}$. $-\partial(\varepsilon(t)^2)/\partial w_j$ was calculated as follows:

$$\begin{aligned}
&-\partial(\varepsilon(t)^2)/\partial w_j \\
&= -2\varepsilon(t)\partial\varepsilon(t)/\partial w_j \\
&= -2\varepsilon(t)\partial(v_{true}(t) - \boldsymbol{w}^{\mathrm{T}}\, \boldsymbol{x}(t))/\partial w_j \\
&= -2\varepsilon(t)(-\,x_j(t)) \\
&= 2\varepsilon(t)x_j(t) \\
&\approx 2\delta(t)x_j(t) \quad \text{(Eq. 1.5)}
\end{aligned}$$

In the last line, since $\varepsilon(t)$ was unavailable as $v_{true}(t)$ was unknown, it was approximated by the TD-RPE:

$$\delta(t) = r(t) + \gamma v(t+1) - v(t). \quad \text{(Eq. 1.6)}$$

$-\partial(\varepsilon(t)^2)/\partial A_{ij}$ was calculated as follows:

$$\begin{aligned}
&-\partial(\varepsilon(t)^2)/\partial A_{ij} \\
&= -2\varepsilon(t)\partial(v_{true}(t) - \boldsymbol{w}^{\mathrm{T}}\, \boldsymbol{x}(t))/\partial A_{ij} \\
&\approx 2\delta(t)\partial(\boldsymbol{w}^{\mathrm{T}}\, \boldsymbol{f}(\mathbf{A}\boldsymbol{x}(t-1) + \mathbf{B}\boldsymbol{o}(t-1)))/\partial A_{ij} \\
&= 2\delta(t)x_j(t-1)(0.5 + x_i(t))(0.5 - x_i(t))\,w_i \quad \text{(Eq. 1.7)}
\end{aligned}$$

Similarly, $-\partial(\varepsilon(t)^2)/\partial B_{ik}$was calculated as follows:

$$-\partial(\varepsilon(t)^2)/\partial B_{ik} \approx 2\delta(t)\,o_k(t-1)\,(0.5 + x_i(t))\,(0.5 - x_i(t))\,w_i \quad \text{(Eq. 1.8)}$$

According to these, the online update rule for the value-RNN was determined as follows:

$$w_j \leftarrow w_j + a_{\text{value}}\delta(t)\,x_j(t) \quad \text{(Eq. 1.9)}$$

$$A_{ij} \leftarrow A_{ij} + a_{\text{RNN}}\delta(t)\,x_j(t-1)\,(0.5 + x_i(t))\,(0.5 - x_i(t))\,w_i \quad \text{(Eq. 1.10)}$$

$$B_{ik} \leftarrow B_{ik} + a_{\text{RNN}}\delta(t)\,o_k(t-1)\,(0.5 + x_i(t))\,(0.5 - x_i(t))\,w_i, \quad \text{(Eq. 1.11)}$$

where a_{value} and a_{RNN} were the learning rates. In each simulation, the elements of **A** and **B**, as well as the elements of $\boldsymbol{x}$, were initialized to pseudo standard normal random numbers, and the elements of $\boldsymbol{w}$ were initialized to 0.

Online value-RNN with fixed random feedback (oVRNNrf)

We considered an implementation of the online value-RNN described above in the cortico-basal ganglia-DA system (**Fig. 1**):

$\boldsymbol{x}$: activities of neurons in a cortical region with rich recurrent connections
A : recurrent connection strengths among $\boldsymbol{x}$
$\boldsymbol{o}$: activities of neurons in a cortical region processing sensory inputs
B : feed-forward connection strengths from $\boldsymbol{o}$ to $\boldsymbol{x}$
f : sigmoidal relationship from the input to the output of the cortical neurons
$\boldsymbol{w}$: connection strengths from cortical neurons $\boldsymbol{x}$ to a group of striatal neurons
v : activity of the group of striatal neurons
δ : activity of a group of DA neurons / released DA

$$w_j \leftarrow w_j + a_{\text{value}}\delta(t)\,x_j(t) \quad \text{(Eq. 1.9)}$$

could be naturally implemented as cortico-striatal synaptic plasticity, which depends on DA ($\delta(t)$) and pre-synaptic (cortical) neuronal activity ($x_j(t)$). However, an issue emerged in implementation of the update rules for **A** and **B**:

$$A_{ij} \leftarrow A_{ij} + a_{\text{RNN}}\delta(t)\,x_j(t-1)\,(0.5 + x_i(t))\,(0.5 - x_i(t))\,w_i \quad \text{(Eq. 1.10)}$$

$$B_{ik} \leftarrow B_{ik} + a_{\text{RNN}}\delta(t)\,o_k(t-1)\,(0.5 + x_i(t))\,(0.5 - x_i(t))\,w_i, \quad \text{(Eq. 1.11)}$$

Specifically, w_i included in the rightmost of these update rules (for the strengths of cortico-cortical synapses $A_{ij}$and B_{ik}) is the connection strength from cortical neuron $x_i$to striatal neurons, i.e., the strength of the cortico-striatal synapses (located within the striatum), which is considered to be unavailable at the cortico-cortical synapses (located within the cortex).

As mentioned in the Introduction, this is an example of the long-standing difficulty in biological implementation of backprop, and recently a potential solution for this difficulty, i.e., replacement of the downstream connection strengths in the update rule for upstream connections with fixed random strengths, has been demonstrated in supervised learning of feed-forward and recurrent networks [34, 36,45]. The online value-RNN, which we considered here, differed from supervised learning considered in these previous studies in two ways: i) it was TD learning, apparent in the approximation of the true error $\varepsilon(t)$ by the TD-RPE $\delta(t)$ in the derivation described above, and ii) it used a scalar error (TD-RPE) rather than a vector error. But we expected that the feedback alignment mechanism could still work at least to some extent, and explored it in this study. Specifically, we examined a modified online value-RNN with fixed random feedback (oVRNNrf), in which the update rules for **A** and **B** were modified as follows:

$$A_{ij} \leftarrow A_{ij} + a_{\mathrm{RNN}}\delta(t)\,x_j(t\!-\!1)\,(0.5 + x_i(t))\,(0.5 - x_i(t))\,c_i \quad \text{(Eq. 2.1)}$$

$$B_{ik} \leftarrow B_{ik} + a_{\mathrm{RNN}}\delta(t)\,o_k(t\!-\!1)\,(0.5 + x_i(t))\,(0.5 - x_i(t))\,c_i, \quad \text{(Eq. 2.2)}$$

where w_i in the update rules of the online value-RNN with backprop (oVRNNbp) was replaced with a fixed random parameter c_i. Notably, these modified update rules for the cortico-cortical connections **A** and **B** required only pre-synaptic activities ($x_j(t-1)$, $o_k(t-1)$), post-synaptic activities ($x_i(t)$), TD-RPE-representing DA ($\delta(t)$), and fixed random strengths (c_i), which would all be available at the cortico-cortical synapses given that VTA DA neurons project not only to the striatum but also to the cortex and random c_i could be provided by intrinsic heterogeneity. In each simulation, the elements of $\boldsymbol{c}$ were initialized to pseudo standard normal random numbers.

Revised online value-RNN models with further biological constraints

In the later part of this study, we examined revised online value-RNN models with further biological constraints. Specifically, we considered models, in which the value weights and the activities of neurons in the RNN were constrained to be non-negative. In order to do so, the update rule for $\boldsymbol{w}$ was modified to:

$$w_j \leftarrow \max(0,\ w_j + a_{\mathrm{value}}\delta(t)\,x_j(t)), \quad \text{(Eq. 3.1)}$$

where $\max(q_1, q_2)$ returned the maximum of q_1 and q_2. Also, the sigmoidal input-output function was replaced with

$$f(z) = 1/(1 + \exp(-z)), \quad \text{(Eq. 3.2)}$$

and the elements of $\boldsymbol{x}$ were initialized to pseudo uniform [0 1] random numbers. The backprop-based update rules for **A** and **B** in oVRNNbp were replaced with

$$A_{ij} \leftarrow A_{ij} + a_{\mathrm{RNN}}\delta(t)\,x_j(t\!-\!1)\,x_i(t)\,(1 - x_i(t))\,w_i \quad \text{(Eq. 3.3)}$$

$$B_{ik} \leftarrow B_{ik} + a_{\mathrm{RNN}}\delta(t)\,o_k(t\!-\!1)\,x_i(t)\,(1 - x_i(t))\,w_i. \quad \text{(Eq. 3.4)}$$

We referred to the model with these modifications to oVRNNbp as oVRNNbp-rev.

As a revised online value-RNN with fixed random feedback (oVRNNrf), in addition to the abovementioned modifications of the update of $\boldsymbol{w}$, the sigmoidal input-output function, and the initialization of $\boldsymbol{x}$, the fixed random feedback $\boldsymbol{c}$ was assumed to be non-negative. Specifically, the elements of $\boldsymbol{c}$ were set to pseudo uniform [0 1] random numbers. Moreover, the update rules for **A** and **B** were replaced with

(when $x_i(t) \le 0.5$)

$$A_{ij} \leftarrow A_{ij} + a_{\mathrm{RNN}}\delta(t)\,x_j(t\!-\!1)\,x_i(t)\,(1 - x_i(t))\,c_i \quad \text{(Eq. 3.5)}$$

$$B_{ik} \leftarrow B_{ik} + a_{\mathrm{RNN}}\delta(t)\,o_k(t\!-\!1)\,x_i(t)\,(1 - x_i(t))\,c_i. \quad \text{(Eq. 3.6)}$$

(when $x_i(t) > 0.5$)

$$A_{ij} \leftarrow A_{ij} + 0.25 a_{\mathrm{RNN}} \delta(t) x_j(t\text{–}1) c_i \quad \text{(Eq. 3.7)}$$

$$B_{ik} \leftarrow B_{ik} + 0.25 a_{\mathrm{RNN}} \delta(t) o_k(t\text{–}1) c_i. \quad \text{(Eq. 3.8)}$$

so that the originally non-monotonic dependence on $x_i(t)$ (post-synaptic activity) became monotonic + saturation (**Fig. 5B**). These update rules with non-negative c_i could be said to be Hebbian with additional modulation by TD-RPE (Hebbian under positive TD-RPE). We referred to the model with these modifications to oVRNNrf as oVRNNrf-bio. In the right panels of **Fig. 6E,F**, we also examined the model where the modified update rules of oVRNNrf-bio were changed back to the original ones, referred to as oVRNNrf-rev. In some simulations in **Fig. 8E-G**, we examined a modified oVRNNrf-bio with a slight decay (forgetting) of value weights, in which each element of $\boldsymbol{w}$ decayed at every time-step:

$$w_i \leftarrow (1 - dr) w_i, \quad \text{(Eq. 3.9)}$$

where dr was the decay rate per time-step and was set to 0.001 or 0.002.

We further examined extensions of oVRNNbp-rev and oVRNNrf-bio, referred to as oVRNNbp-rev-ei and oVRNNrf-bio-ei, which incorporated excitatory E-units and inhibitory I-units (**Fig. 9A**). Based on biological suggestions (see the Results), we made the following assumptions. Each E-unit received inputs from the observation units $\boldsymbol{o}$ (connections: $\mathbf{B_E}$), all the E-units (connections: $\mathbf{A_E}$), and a particular I-unit (with a strength h), and projected to the striatal value unit (connections: $\boldsymbol{w}$). Each I-unit received inputs from the observation units $\boldsymbol{o}$ (connections: $\mathbf{B_I}$) and all the E-units (connections:

$\mathbf{A_I}$). Excitation from the observation units and E-units to E- and I-units took one time-step whereas I→E inhibition operated within a time-step. The activation function for E-unit and plasticity rules for connections from/to E-units were the same as those for the RNN unit in the original models. I-unit had a linear activation function, and there was no plasticity for connections from/to I-units. Update rule for $\boldsymbol{w}$ was the same as the original one with the activity of the RNN units replaced with the activity of E-units. Equations for the activities of E-units and I-units, $\boldsymbol{x}_\mathbf{E}$ and $\boldsymbol{x}_\mathbf{I}$, are given as follows:

$$\boldsymbol{x}_\mathbf{I}(t+1) = \mathbf{A_I}\boldsymbol{x}_\mathbf{E}(t) + \mathbf{B_I}\boldsymbol{o}(t) \quad \text{(Eq. 3.10)}$$

$$\boldsymbol{x}_\mathbf{E}(t+1) = \boldsymbol{f}(\mathbf{A_E}\boldsymbol{x}(t) + \mathbf{B_E}\boldsymbol{o}(t) - h\boldsymbol{x}_\mathbf{I}(t+1)), \quad \text{(Eq. 3.11)}$$

where h was set to 1. The elements of $\mathbf{A_I}$, $\mathbf{B_I}$, $\mathbf{A_E}$, and $\mathbf{B_E}$ were initialized to be non-negative:

$$\max(0, 3 + z),$$

where z was a pseudo standard normal random number. The elements of $\boldsymbol{x}_\mathbf{E}$ were initialized to pseudo uniform [0 1] random numbers, and the initial values of $\boldsymbol{x}_\mathbf{I}$were determined according to the abovementioned equation.

Incorporation of action selection

We considered extensions of oVRNNbp-rev and oVRNNrf-bio that incorporated an actor-critic architecture, referred to as oVRNNbp-rev-ac and oVRNNrf-bio-ac (**Fig. 11A**). Each RNN unit additionally connected to additional two units representing the action-values of action 1 and action 2 (q_1 and q_2):

$$\boldsymbol{q}(t) = \mathbf{U}\boldsymbol{x}(t), \quad \text{(Eq. 4.1)}$$

where $\mathbf{U} = (u_{kj})$ consisted of two row vectors that represented the preferences of the two actions. At the time-step next to cue-presentation, one action was selected in a soft-max manner based on the action values. Specifically, action k was selected with the probability of

$$\exp(\beta q_k)/(\exp(\beta q_1) + \exp(\beta q_2)), \quad \text{(Eq. 4.2)}$$

where β was the inverse temperature parameter, set to 1 or 2, representing the degree of exploitation over exploration. Selected action was then informed to the RNN units. Specifically, the observation layer had two additional elements (***o'*** in **Fig. 11A**) corresponding to the two actions. These elements became 1 when the corresponding action was selected and 0 otherwise. The preference of selected action k was updated by using the TD-RPE $\delta(t)$ as follows:

$$u_{kj} \leftarrow u_{kj} + a_{\text{pref}}\delta(t)\,x_j(t), \quad \text{(Eq. 4.3)}$$

where a_{pref} was the learning rate. In order to prevent unbounded increase of action preference, we assumed a slight decay of all the action preferences at every time-step:

$$u_{kj} \leftarrow (1 - dr)\,u_{kj}, \quad \text{(Eq. 4.4)}$$

where dr was the decay rate per time-step and was set to 0.001. u_{kj} (i.e., the elements of **U**) were initialized to 0. The connection weights from the action-observation units ***o'***, as well as from ***o*** and the RNN units, onto the RNN units were initialized to pseudo standard normal random numbers.

Simulation of the tasks

In the Pavlovian cue-reward association task, at time 1 of each trial, cue observation was received by the RNN, and at time 4, reward observation was received. Trial was pseudo-randomly ended at time 7, 8, 9, or 10, and the next trial started from the next time-step (i.e., inter-trial interval (ITI) was 4, 5, 6, or 7 time-steps with equal probabilities). Reward size was $r = 1$. We also conducted simulations with longer cue-reward delays, in which reward was given at time 5, 6, or 7, and the end of trial was shifted accordingly. The tasks with probabilistic structures (task 1 and task 2) were implemented in the same way except that reward timing was not time 4 but time 3 or 5 with equal probabilities, specifically, 50% and 50% in task 1 and 30% and 30% in task 2, and there was no reward in the remaining 40% of trials in task 2. The tasks with action selection were also implemented in the same way except that size 2 reward was received, i.e., the reward term in the TD-RPE calculation as well as the reward-corresponding element of the observation inputs were set to 2, at time 4 when action 1 was selected whereas size 1 reward was received at time 4 (in the first choice task) or time 3 (in the second choice task) when action 2 was selected.

The cue or reward state/timing, mentioned in the text and marked in the figures, was defined to be the timing when the RNN received the cue or reward observation, respectively. Specifically, if $\boldsymbol{o}(t) = (1\ 0)^{\mathrm{T}}$ or $\boldsymbol{o}(t) = (0\ 1)^{\mathrm{T}}$ at time t, $t + 1$ was defined to be a cue or reward timing, respectively. For the agents with punctate state representation, which is also referred to as the complete serial compound (CSC) representation [1, 48, 133], each timing from a cue in the tasks was represented by a 10-dimensional one-hot vector, starting from $(1\ 0\ 0\ \dots\ 0)^{\mathrm{T}}$ for the cue state, with the next state $(0\ 1\ 0\ \dots\ 0)^{\mathrm{T}}$ and so on.

In the simulations of the cue-reward association task with distractor cue, the observation units $\boldsymbol{o} = (o_k)$ had an additional element o_3 (**Fig. 10A**), which was 1 at the time-steps where distractor cue was present and 0 otherwise. We examined four cases, in which the probability of the presence of distractor cue at every time-step throughout the task was 0, 0.1, 0.2, and 0.3 (**Fig. 10B-E**).

Learning rates were set as follows. For the models with punctate state representation, $a_{\text{value}} = 0.1$. For oVRNNbp, oVRNNrf, and the model with untrained RNN compared with these two, $a_{\text{RNN}} = 0.1$ and $a_{\text{value}} = 0.1/(n/7)$. For oVRNNbp-rev, oVRNNrf-bio, and the models with untrained RNN compared with these two, a_{RNN} = 0.1 and a_{value} = a_{pref} = 0.1/(n/12) except for the right panels of **Fig. 6J**, and a_{RNN} = 0.05 and a_{value} = 0.05/(n/12) for the right panels of **Fig. 6J**. Time discount factor (γ) was set to 0.8.

Estimation of true state/timing values

As for the Pavlovian cue-reward association task, we defined states after agent's receival of cue information by relative timings from the cue, and estimated their (true) values by simulations according to the definition of state value. We generated a sequence of cues and rewards corresponding to 1000 trials with the ITI after the first trial, ITI_1, fixed to one of the possible lengths (4, 5, 6, or 7 time-steps), and calculated cumulative discounted future rewards within the sequence:

$$\Sigma_{t\_\mathrm{rew}}(r\gamma^{t\_\mathrm{rew}}),$$

where $_{t\_\mathrm{rew}}$ denotes the time-step of each reward counted from the starting state, starting from +1, ..., and +3 + ITI_1 time steps from a cue (the last one corresponded to the cue timing of the next trial). For each case where ITI_1 = 4, 5, 6, or 7, we repeated this 1000 times, generating 1000 sequences (i.e., 1000 simulations of 1000 trials), with different sets of pseudo-random numbers, and calculated the average over these 1000 sequences (we refer to these as ITI_1-specific values). We estimated the value of each state of +1, ..., and +7 time steps from cue (i.e., −2, ..., +4 time-steps from reward) by taking the average of the ITI_1-specific values for four possible ITI_1.

We also estimated the true values of the cue timing and one and two timing(s) before it in the following way; these values could not be estimated in the abovementioned way because agent should not know the length of ITI (i.e., when ITI ends) until receiving cue information at the cue timing. In the case where ITI is in fact 4 time-steps, until receiving the next cue, agent should think that ITI can be 4, 5, 6, or 7 time-steps with equal probabilities (1/4 for each). Thus, the value of next cue timing and one and two timing(s) before it should be the average of the four ITI_1-specific values of +4, +3, and +2 time-steps from reward. Similarly, in the case where ITI is in fact 5 time-steps, until the previous time-step of the next cue, agent should think that ITI can be 4, 5, 6, or 7 time-steps with equal probabilities (1/4 for each). Thus, the value of one and two timing(s) before next cue should be the average of the four ITI_1-specific values of +4 and +3 time-steps from reward. On the other hand, at the timing of the next cue, agent should think that ITI can be 5, 6, or 7 (but not 4) time-steps with equal probabilities (1/3 for each). Thus, the value of next cue timing should be the average of the three ITI_1-specific values (for ITI_1 = 5, 6, or 7) of +5 time-steps from reward. Similar considerations can be made for the cases where ITI is in fact 6 or 7 time-steps. And then, the "true" value of (next) cue timing can be calculated as the average of the values of next cue timing in the cases where ITI is in fact 4, 5, 6, or 7 time-steps. Using these estimated true state values, we calculated TD-RPE at each state/timing (−2, −1, ..., and +5 time steps from cue). True state/timing values in the cases where the cue-reward delay was 4, 5, or 6 time-steps were estimated in the same way.

We also estimated true state/timing values for tasks 1 and 2 that had probabilistic structures. As for task 1, we first estimated the values of each timing in each of the trial types (**Fig. 4Ba**, left), in which reward was given at early (2 time-steps after cue) or late (4 time-steps after cue) timing, in the same manner (but using 10000 rather than 1000 simulations for each condition) as done for the cue- reward association task mentioned above (values of the cue timing and the one and two timing(s) before cue after each trial type were also estimated). Then, based on the agent's belief about trial types (**Fig. 4Bb**, left), we defined the following states: +1 and +2 time steps from cue (i.e., states visited (entered) before knowing whether reward was given at the early timing (= +2 time step from cue)), +3, 4, 5, and 6 time steps from cue after reception of reward at the early timing, and +3, 4, 5, and 6 time steps from cue after no reception of reward at the early timing (**Fig. 4Bc**, left-top). We calculated the true values of these states and also of the cue timing and one and two timing(s) before cue (**Fig. 4Bc**, left-bottom) by taking (mathematical) expected value of the abovementioned estimated value of each timing in each trial type. Using these true values, we calculated TD-RPE (**Fig. 4C**, left).

As for task 2, we first estimated the values of each timing in each of the trial types (**Fig. 4Ba**, right), in which reward was given at early (2 time-steps after cue) or late (4 time-steps after cue) timing or was not given. Then, based on the agent's belief about trial types (**Fig. 4Bb**, right), we defined the following states: +1 and +2 time steps from cue (i.e., states visited (entered) before knowing whether reward was given at the early timing), +3, 4, 5, and 6 time steps from cue after reception of reward at the early timing, +3 and 4 time steps from cue after no reception of reward at the early timing (states visited (entered) before knowing whether reward was given at the late timing (= +4 time step from cue)), +5 and 6 time steps from cue after reception of reward at the late timing, and +5 and 6 time steps from cue after no reception of reward at both early and late timings (**Fig. 4Bc**, right-top). We estimated the true values of these states and also of the cue timing and one and two timing(s) before cue (**Fig. 4Bc**, right-bottom) in the same manner as for task 1, and using these true values, we calculated TD-RPE (**Fig. 4C**, right).

Analyses, software, and code availability

SEM (Standard error of the mean) was approximated by SD (standard deviation)/$\sqrt{N}$ (number of samples). Cohen's d using an average variance was calculated as (difference in the means) / (square root of the average of variances). Linear regression, principal component analysis (PCA), Wilcoxon rank sum test, and t-tests were conducted by using R (functions lm, prcomp, wilcox.exact (in package exactRankTests), and t.test). Difference in Wilcoxon rank sum test and t-tests was reported when $p < 0.05$. Simulations were conducted by using MATLAB, and pseudo-random numbers were implemented by using rand, randn, and randperm functions. The codes for simulations and analyses are available at GitHub (*https://github.com/kenjimoritagithub/oVRNN1*).

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Additional information

Author contributions

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Author information

Takayuki Tsurumi

Physical and Health Education, Graduate School of Education, The University of Tokyo, Tokyo, Japan, Theoretical Sciences Visiting Program, Okinawa Institute of Science and Technology, Okinawa, Japan

Ayaka Kato

Theoretical Sciences Visiting Program, Okinawa Institute of Science and Technology, Okinawa, Japan, Department of Psychiatry, Icahn School of Medicine at Mount Sinai, New York, United States
ORCID iD: 0000-0002-6306-6600

Arvind Kumar

Theoretical Sciences Visiting Program, Okinawa Institute of Science and Technology, Okinawa, Japan, Division of Computational Science and Technology, School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology, Stockholm, Sweden
ORCID iD: 0000-0002-8044-9195

Kenji Morita
Physical and Health Education, Graduate School of Education, The University of Tokyo, Tokyo, Japan, Theoretical Sciences Visiting Program, Okinawa Institute of Science and Technology, Okinawa, Japan, International Research Center for Neurointelligence (WPI-IRCN), The University of Tokyo, Tokyo, Japan
ORCID iD: 0000-0003-2192-4248

For correspondence: morita@p.u-tokyo.ac.jp

Editors

Reviewing Editor
Richard Naud
University of Ottawa, Ottawa, Canada

Senior Editor
Panayiota Poirazi
FORTH Institute of Molecular Biology and Biotechnology, Heraklion, Greece

Reviewer #1 (Public review):

Summary:

Can a plastic RNN serve as a basis function for learning to estimate value. In previous work this was shown to be the case, with a similar architecture to that proposed here. The learning rule in previous work was back-prop with an objective function that was the TD error function (delta) squared. Such a learning rule is non-local as the changes in weights within the RNN, and from inputs to the RNN depends on the weights from the RNN to the output, which estimates value. This is non-local, and in addition, these weights themselves change over learning. The main idea in this paper is to examine if replacing the values of these non-local changing weights, used for credit assignment, with random fixed weights can still produce similar results to those obtained with complete bp. This random feedback approach is motivated by a similar approach used for deep feed-forward neural networks.

This work shows that this random feedback in credit assignment performs well but is not as well as the precise gradient-based approach. When more constraints due to biological plausibility are imposed performance degrades. These results are consistent with previous results on random feedback.

Strengths:

• The authors show that random feedback can approximate well a model trained with detailed credit assignment.
• The authors simulate several experiments including some with probabilistic reward schedules and show results similar to those obtained with detailed credit assignments as well as in experiments.
• The paper examines the impact of more biologically realistic learning rules and the results are still quite similar to the detailed back-prop model.

Weaknesses:

• The impact of the article is limited by using a network with discrete time-steps, and only a small number of time steps from stimulus to reward. They assume that each time step is on the order of hundreds of ms. They justify this by pointing to some slow intrinsic mechanisms, but they do not implement these slow mechanisms is a network with short time steps, instead they assume without demonstration that these could work as suggested. This is a reasonable first approximation, but its validity should be explicitly tested.

• As the delay between cue and reward increases the performance decreases. This is not surprising given the proposed mechanism, but is still a limitation, especially given that we do not really know what a is the reasonable value of a single time step.

https://doi.org/10.7554/eLife.104101.2.sa3

Reviewer #2 (Public review):

Summary:

Tsurumi et al. show that recurrent neural networks can learn state and value representations in simple reinforcement learning tasks when trained with random feedback weights. The traditional method of learning for recurrent network in such tasks (backpropogation through time) requires feedback weights which are a transposed copy of the feed-forward weights, a biologically implausible assumption. This manuscript builds on previous work regarding "random feedback alignment" and "value-RNNs", and extends them to a reinforcement learning context. The authors also demonstrate that certain non-negative constraints can enforce a "loose alignment" of feedback weights. The author's results suggest that random feedback may be a powerful tool of learning in biological networks, even in reinforcement learning tasks.

Strengths:

The authors describe well the issues regarding biologically plausible learning in recurrent networks and in reinforcement learning tasks. They take care to propose networks which might be implemented in biological systems and compare their proposed learning rules to those already existing in literature. Further, they use small networks on relatively simple tasks, which allows for easier intuition into the learning dynamics.

Weaknesses:

The principles discovered by the authors in these smaller networks are not applied to larger networks or more complicated tasks with long temporal delays (>100 timesteps), so it remains unclear to what degree these methods can scale or can be used more generally.

Comments on revisions: I would still want to see how well the network learns tasks with longer time delays (on the order of 100 or even 1000 timesteps). Previous work has shown that random feedback struggles to encode longer timescales (see Murray 2019, Figure 2), so I would be interested to see how that translates to the RL context in your model.

https://doi.org/10.7554/eLife.104101.2.sa2

Reviewer #3 (Public review):

Summary:

The paper studies learning rules in a simple sigmoidal recurrent neural network setting. The recurrent network has a single layer of 10 to 40 units. It is first confirmed that feedback alignment (FA) can learn a value function in this setting. Then so-called bio-plausible constraints are added: (1) when value weights (readout) is non-negative, (2) when the activity is non-negative (normal sigmoid rather than downscaled between -0.5 and 0.5), (3) when the feedback weights are non-negative, (4) when the learning rule is revised to be monotic: the weights are not downregulated. In the simple task considered all four biological features do not appear to impair totally the learning.

Strengths:

(1) The learning rules are implemented in a low-level fashion of the form: (pre-synaptic-activity) x (post-synaptic-activity) x feedback x RPE. Which is therefore interpretable in terms of measurable quantities in the wet-lab.

(2) I find that non-negative FA (FA with non negative c and w) is the most valuable theoretical insight of this paper: I understand why the alignment between w and c is automatically better at initialization.

(3) The task choice is relevant, since it connects with experimental settings of reward conditioning with possible plasticity measurements.

Weaknesses:

(4) The task is rather easy, so it's not clear that it really captures the computational gap that exists with FA (gradient-like learning) and simpler learning rule like a delta rule: RPE x (pre-synpatic) x (post-synaptic). To control if the task is not too trivial, I suggest adding a control where the vector c is constant c_i=1.

(5) Related to point 3), the main strength of this paper is to draw potential connection with experimental data. It would be good to highlight more concretely the prediction of the theory for experimental findings. (Ideally, what should be observed with non-negative FA that is not expected with FA or a delta rule (constant global feedback) ?).

(6a) Random feedback with RNN in RL have been studied in the past, so it is maybe worth giving some insights how the results and the analyzes compare to this previous line of work (for instance in this paper [1]). For instance, I am not very surprised that FA also works for value prediction with TD error. It is also expected from the literature that the RL + RNN + FA setting would scale to tasks that are more complex than the conditioning problem proposed here, so is there a more specific take-home message about non-negative FA? or benefits from this simpler toy task?

(6b) Related to task complexity, it is not clear to me if non-negative value and feedback weights would generally scale to harder tasks. If the task in so simple that a global RPE signal is sufficient to learn (see 4 and 5), then it could be good to extend the task to find a substantial gap between: global RPE, non-negative FA, FA, BP. For a well chosen task, I expect to see a performance gap between any pair of these four learning rules. In the context of the present paper, this would be particularly interesting to study the failure mode of non-negative FA and the cases where it does perform as well as FA.

(7) I find that the writing could be improved, it mostly feels more technical and difficult than it should. Here are some recommendations:
7a) For instance, the technical description of the task (CSC) is not fully described and requires background knowledge from other paper which is not desirable.
7b) Also the rationale for the added difficulty with the stochastic reward and new state is not well explained.
7c) In the technical description of the results I find that the text dives into descriptive comments of the figures but high-level take home messages would be helpful to guide the reader. I got a bit lost, although I feel that there is probably a lot of depth in these paragraphs.

(8) Related to the writing issue and 5), I wished that "bio-plausibility" was not the only reason to study positive feedback and value weights. Is it possible to develop a bit more specifically what and why this positivity is interesting? Is there an expected finding with non-negative FA

both in the model capability? or maybe there is a simpler and crisp take-home message to communicate the experimental predictions to the community would be useful?

[1] https://www.nature.com/articles/s41467-020-17236-y

Comments on revisions:

Thank you for addressing all my comments in your reply.

https://doi.org/10.7554/eLife.104101.2.sa1

Author response:

The following is the authors' response to the original reviews

Summary of our revisions

(1) We have explained the reason why the untrained RNN with readout (value-weight) learning only could not well learn the simple task: it is because we trained the models continuously across trials with random inter-trial intervals rather than separately for each episodic trial and so it was not trivial for the models to recognize that cue presentation in different trials constitutes a same single state since the activities of untrained RNN upon cue presentation should differ from trial to trial (Line 177-185).

(2) We have shown that dimensionality was higher in the value-RNNs than in the untrained RNN (Fig. 2K,6H).

(3) We have shown that even when distractor cue was introduced, the value-RNNs could learn the task (Fig. 10).

(4) We have shown that extended value-RNNs incorporating excitatory and inhibitory units and conforming to the Dale's law could still learn the tasks (Fig. 9,10-right column).

(5) In the original manuscript, the non-negatively constrained value-RNN showed loose alignment of value-weight and random feedback from the beginning but did not show further alignment over trials. We have clarified its reason and found a way, introducing a slight decay (forgetting), to make further alignment occur (Fig. 8E,F).

(6) We have shown that the value-RNNs could learn the tasks with longer cue-reward delay (Fig. 2M,6J) or action selection (Fig. 11), and found cases where random feedback performed worse than symmetric feedback.

(7) We compared our value-RNNs with e-prop (Bellec et al., 2020, Nat Commun). While e-prop incorporates the effects of changes in RNN weights across distant times through "eligibility trace", our value-RNNs do not. The reason why our models can still learn the tasks with cue-reward delay is considered to be because our models use TD error and TD learning itself, even TD(0) without eligibility trace, is a solution for temporal credit assignment. In fact, TD error-based e-prop was also examined, but for that, result with symmetric feedback, but not with random feedback, was shown (their Fig. 4,5) while for another setup of reward-based e-prop without TD error, result with random feedback was shown (their SuppFig. 5). We have noted these in Line 695-711 (and also partly in Line 96-99).

(8) In the original manuscript, we emphasized only the spatial locality (random rather than symmetric feedback) of our learning rule. But we have now also emphasized the temporal locality (online learning) as it is also crucial for bio-plausibility and critically different from the original value-RNN with BPTT. We also changed the title.

(9) We have realized that our estimation of true state values was invalid (as detailed in page 34 of this document). Effects of this error on performance comparisons were small, but we apologize for this error.

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *Can a plastic RNN serve as a basis function for learning to estimate value. In previous work this was shown to be the case, with a similar architecture to that proposed here. The learning rule in previous work was back-prop with an objective function that was the TD error function (delta) squared. Such a learning rule is non-local as the changes in weights within the RNN, and from inputs to the RNN depends on the weights from the RNN to the output, which estimates value. This is non-local, and in addition, these weights themselves change over learning. The main idea in this paper is to examine if replacing the values of these non-local changing weights, used for credit assignment, with random fixed weights can still produce similar results to those obtained with complete bp. This random feedback approach is motivated by a similar approach used for deep feed-forward neural networks.*
>
> *This work shows that this random feedback in credit assignment performs well but is not as well as the precise gradient-based approach. When more constraints due to biological plausibility are imposed performance degrades. These results are not surprising given previous results on random feedback. This work is incomplete because the delay times used were only a few time steps, and it is not clear how well random feedback would operate with longer delays. Additionally, the examples simulated with a single cue and a single reward are overly simplistic and the field should move beyond these exceptionally simple examples.*
>
> *Strengths:*
>
> *• The authors show that random feedback can approximate well a model trained with detailed credit assignment.*
>
> *• The authors simulate several experiments including some with probabilistic reward schedules and show results similar to those obtained with detailed credit assignments as well as in experiments.*
>
> *• The paper examines the impact of more biologically realistic learning rules and the results are still quite similar to the detailed back-prop model.*
>
> *Weaknesses:*
>
> **please note that we numbered your public review comments and recommendations for the authors as Pub1 and Rec1 etc so that we can refer to them in our replies to other comments.*
>
> *Pub1. The authors also show that an untrained RNN does not perform as well as the trained RNN. However, they never explain what they mean by an untrained RNN. It should be clearly explained.*
>
> *These results are actually surprising. An untrained RNN with enough units and sufficiently large variance of recurrent weights can have a high-dimensionality and generate a complete or nearly complete basis, though not orthonormal (e.g: Rajan&Abbott 2006). It should be possible to use such a basis to learn this simple classical conditioning paradigm. It would be useful to measure the dimensionality of network dynamics, in both trained and untrained RNN's.*

We have added an explanation of untrained RNN in Line 144-147:

"As a negative control, we also conducted simulations in which these connections were not updated from initial values, referring to as the case with "untrained (fixed) RNN". Notably, the value weights w (i.e., connection weights from the RNN to the striatal value unit) were still trained in the models with untrained RNN."

We have also analyzed the dimensionality of network dynamic by calculating the contribution ratios of each principal component of the trajectory of RNN activities. It was revealed that the contribution ratios of later principal components were smaller in the cases with untrained RNN than in the cases with trained value RNN. We have added these results in Fig. 2K and Line 210-220 (for our original models without non-negative constraint):

"In order to examine the dimensionality of RNN dynamics, we conducted principal component analysis (PCA) of the time series (for 1000 trials) of RNN activities and calculated the contribution ratios of PCs in the cases of oVRNNbp, oVRNNrf, and untrained RNN with 20 RNN units. Figure 2K shows a log of contribution ratios of 20 PCs in each case. Compared with the case of untrained RNN, in oVRNNbp and oVRNNrf, initial component(s) had smaller contributions (PC1 (t-test $p = 0.00018$ in oVRNNbp; $p = 0.0058$ in oVRNNrf) and PC2 ($p = 0.080$ in oVRNNbp; $p = 0.0026$ in oVRNNrf)) while later components had larger contributions (PC3~10,15~20 $p < 0.041$ in oVRNNbp; PC5~20 $p < 0.0017$ in oVRNNrf) on average, and this is considered to underlie their superior learning performance. We noticed that late components had larger contributions in oVRNNrf than in oVRNNbp, although these two models with 20 RNN units were comparable in terms of cue~reward state values (Fig. 2J-left)."

and Fig. 6H and Line 412-416 (for our extended models with non-negative constraint):

"Figure 6H shows contribution ratios of PCs of the time series of RNN activities in each model with 20 RNN units. Compared with the cases with naive/shuffled untrained RNN, in oVRNNbp-rev and oVRNNrf-bio, later components had relatively high contributions (PC5~20 p < 1.4×10,sup>−6 (t-test vs naive) or < 0.014 (vs shuffled) in oVRNNbp-rev; PC6~20 $p < 2.0\times10^{-7}$ (vs naive) or PC7~20 $p < 5.9\times10^{-14}$ (vs shuffled) in oVRNNrf-bio), explaining their superior value-learning performance."

Regarding the poor performance of the model with untrained RNN, we would like to add a note. It is sure that untrained RNN with sufficient dimensions should be able to well represent just <10 different states, and state values should be able to be well learned through TD learning regardless of whatever representation is used. However, a difficulty (nontriviality) lies in that because we modeled the tasks in a continuous way, rather than in an episodic way, the activity of untrained RNN upon cue presentation should generally differ from trial to trial. Therefore, it was not trivial for RNN to know that cue presentation in different trials, even after random lengths of inter-trial interval, should constitute a same single state. We have added this note in Line 177-185:

"This inferiority of untrained RNN may sound odd because there were only four states from cue to reward while random RNN with enough units is expected to be able to represent many different states (c.f., [49]) and the effectiveness of training of only the readout weights has been shown in reservoir computing studies [50-53]. However, there was a difficulty stemming from the continuous training across trials (rather than episodic training of separate trials): the activity of untrained RNN upon cue presentation generally differed from trial to trial, and so it is non-trivial that cue presentation in different trials should be regarded as the same single state, even if it could eventually be dealt with at the readout level if the number of units increases."

The original value RNN study (Hennig et al., 2023, PLoS Comput Biol) also modeled tasks in a continuous way (though using backprop-through-time (BPTT) for training) and their model with untrained RNN also showed considerably larger RPE error than the value RNN even when the number of RNN units was 100 (the maximum number plotted in their Fig. 6A).

> *Pub2. The impact of the article is limited by using a network with discrete time-steps, and only a small number of time steps from stimulus to reward. What is the length of each time step? If it's on the order of the membrane time constant, then a few time steps are only tens of ms. In the classical conditioning experiments typical delays are of the order to hundreds of milliseconds to seconds. Authors should test if random feedback weights work as well for larger time spans. This can be done by simply using a much larger number of time steps.*

In the revised manuscript, we examined the cases in which the cue-reward delay (originally 3 time steps) was elongated to 4, 5, or 6 time-steps. Our online value RNN models with random feedback could still achieve better performance (smaller squared value error) than the models with untrained RNN, although the performance degraded as the cue-reward delay increased. We have added these results in Fig. 2M and Line 223-228 (for our original models without non-negative constraint)

“We further examined the cases with longer cue-reward delays. As shown in Fig. 2M, as the delay increased, the mean squared error of state values (at 3000-th trial) increased, but the relative superiority of oVRNNbp and oVRNNrf over the model with untrained RNN remained to hold, except for cases with small number of RNN units (5) and long delay (5 or 6) ($p < 0.0025$ in Wilcoxon rank sum test for oVRNNbp or oVRNNrf vs untrained for each number of RNN units for each delay).”

and Fig. 6J and Line 422-429 (for our extended models with non-negative constraint):

“Figure 6J shows the cases with longer cue-reward delays, with default or halved learning rates. As the delay increased, the mean squared error of state values (at 3000-th trial) increased, but the relative superiority of oVRNNbp-rev and oVRNNrf-bio over the models with untrained RNN remained to hold, except for a few cases with 5 RNN units (5 delay oVRNNrf-bio vs shuffled with default learning rate, 6 delay oVRNNrf-bio vs naive or shuffled with halved learning rate) ($p < 0.047$ in Wilcoxon rank sum test for oVRNNbp-rev or oVRNNrf-bio vs naive or shuffled untrained for each number of RNN units for each delay).”

Also, we have added the note about our assumption and consideration on the time-step that we described in our provisional reply in Line 136-142:

“We assumed that a single RNN unit corresponds to a small population of neurons that intrinsically share inputs and outputs, for genetic or developmental reasons, and the activity of each unit represents the (relative) firing rate of the population. Cortical population activity is suggested to be sustained not only by fast synaptic transmission and spiking but also, even predominantly, by slower synaptic neurochemical dynamics [46] such as short-term facilitation, whose time constant can be around 500 milliseconds [47]. Therefore, we assumed that single time-step of our rate-based (rather than spike-based) model corresponds to 500 milliseconds.”

> *Pub3. In the section with more biologically constrained learning rules, while the output weights are restricted to only be positive (as well as the random feedback weights), the recurrent weights and weights from input to RNN are still bi-polar and can change signs during learning. Why is the constraint imposed only on the output weights? It seems reasonable that the whole setup will fail if the recurrent weights were only positive as in such a case most neurons will have very similar dynamics, and the network dimensionality would be very low. However, it is possible that only negative weights might work. It is unclear to me how to justify that bipolar weights that change sign are appropriate for the recurrent connections and inappropriate for the output connections.*

> *On the other hand, an RNN with excitatory and inhibitory neurons in which weight signs do not change could possibly work.*

We examined extended models that incorporated inhibitory and excitatory units and followed Dale's law with certain assumptions, and found that these models could still learn the tasks. We have added these results in Fig. 9 and subsection “4.1 Models with excitatory and inhibitory units” and described the details of the extended models in Line 844-862:

> *Pub4. Like most papers in the field this work assumes a world composed of a single cue. In the real world there many more cues than rewards, some cues are not associated with any rewards, and some are associated with other rewards or even punishments. In the simplest case, it would be useful to show that this network could actually work if there are additional distractor cues that appear at random either before the CS, or between the CS and US. There are good reasons to believe such distractor cues will be fatal for an untrained RNN, but might work with a trained RNN, either using BPPT or random feedback. Although this assumption is a common flaw in most work in the field, we should no longer ignore these slightly more realistic scenarios.*

We examined the performance of the models in a task in which distractor cue randomly appeared. As a result, our model with random feedback, as well as the model with backprop, could still learn the state values much better than the models with untrained RNN. We have added these results in Fig. 10 and subsection “4.2 Task with distractor cue”

> ***Reviewer #1 (Recommendations for the authors):***
>
> *Detailed comments to authors*
>
> *Rec1. Are the untrained RNNs discussed in methods? It seems quite good in estimating value but has a strong dopamine response at time of reward. Is nothing trained in the untrained RNN or are the W values trained. Untrained RNN are not bad at estimating value, but not as good as the two other options. It would seem reasonable that an untrained RNN (if I understand what it is) will be sufficient for such simple Pavlovian conditioning paradigms. This is provided that the RNN generates a complete, or nearly complete basis. Random RNN's provided that the random weights are chosen properly can indeed generate a nearly complete basis. Once there is a nearly complete temporal basis, it seems that a powerful enough learning rule will be able to learn the very simple Pavlovian conditioning. Since there are only 3 time-steps from cue to reward, an RNN dimensionality of 3 would be sufficient. A failure to get a good approximation can also arise from the failure of the learning algorithm for the output weights (W).*

As we mentioned in our reply to your public comment Pub1 (page 3-5), we have added an explanation of "untrained RNN" (in which the value weights were still learnt) (Line 144-147). We also analyzed the dimensionality of network dynamics by calculating the contribution ratios of principal components of the trajectory of RNN activities, showing that the contribution ratios of later principal components were smaller in the cases with untrained RNN than in the cases with trained value RNN (Fig. 2K/Line 210-220, Fig.6H/Line 412-416). Moreover, also as we mentioned in our reply to your public comment Pub1, we have added a note that even learning of a small number of states was not trivially easy because we considered continuous learning across trials rather than episodic learning of separate trials and thus it was not trivial for the model to know that cue presentation in different trials after random lengths of inter-trial interval should still be regarded as a same single state (Line 177-185).

> *Rec2. For all cases, it will be useful to estimate the dimensionality of the RNN. Is the dimensionality of the untrained RNN smaller than in the trained cases? If this is the case,*

> *this might depend on the choice of the initial random (I assume) recurrent connectivity matrix.*

As mentioned above, we have analyzed the dimensionality of the network dynamics, and as you said, the dimensionality of the model with untrained RNN (which was indeed the initial random matrix as you said, as we mentioned above) was on average smaller than the trained value RNN models (Fig. 2K/Line 210-220, Fig.6H/Line 412-416).

> *Rec3. It is surprising that the error starts increasing for more RNN units above ~15. See discussion. This might indicate a failure to adjust the learning parameters of the network rather than a true and interesting finding.*

Thank you very much for this insightful comment. In the original manuscript, we set the learning rate to a fixed value (0.1), without normalization by the squared norm of feature vector (as we mentioned in Line 656-7 of the original manuscript) because we thought such a normalization could not be locally (biologically) implemented. However, we have realized that the lack of normalization resulted in excessively large learning rate when the number of RNN units was large and it could cause instability and error increase as you suggested. Therefore, in the revised manuscript, we have implemented a normalization of learning rate (of value weights) that does not require non-local computations, specifically, division by the number of RNN units. As a result, the error now monotonically decreased, as the number of RNN units increased, in the non-negatively constrained models (Fig. 6E-left) and also largely in the unconstrained model with random feedback, although still not in the unconstrained model with backprop or untrained RNN (Fig. 2J-left)

> *Rec4. Not numbering equations is a problem. For example, the explanations of feedback alignment (lines 194-206) rely on equations in the methods section which are not numbered. This makes it hard to read these explanations. Indeed, it will also be better to include a detailed derivation of the explanation in these lines in a mathematical appendix. Key equations should be numbered.*

We have added numbers to key equations in the Methods, and references to the numbers of corresponding equations in the main text. Detailed derivations are included in the Methods.

> *Rec5. What is shown in Figure 3C? - an equation will help.*

We have added an explanation using equations in the main text (Line 256-259).

> *Rec6. The explanation of why alignment occurs is not satisfactory, but neither is it in previous work on feedforward networks. The least that should be done though*

Regarding why alignment occurs, what remained mysterious (to us) was that in the case of nonnegatively constrained model, while the angle between value weight vector (w) and the random feedback vector (c) was relatively close (loosely aligned) from the beginning, it appeared (as mentioned in the manuscript) that there was no further alignment over trials, despite that the same mechanism for feedback alignment that we derived for the model without non-negative constraint was expected to operate also under the non-negative constraint. We have now clarified the reason for this, and found a way, introduction of slight decay (forgetting) of value weights, by which feedback alignment came to occur in the non-negatively constraint model. We have added these in the revised manuscript (Line 463-477):

"As mentioned above, while the angle between w and c was on average smaller than 90° from the beginning, there was no further alignment over trials. This seemed mysterious because the mechanism for feedback alignment that we derived for the models without non-negative constraint was expected to work also for the models with non-negative constraint. As a possible reason for the non-occurrence of feedback alignment, we guessed that one or a few element(s) of w grew prominently during learning, and so w became close to an edge or boundary of the non-negative quadrant and thereby angle between w and other vector became generally large (as illustrated in Fig. 8D). Figure 8Ea shows the mean±SEM of the elements of w ordered from the largest to smallest ones after 1500 trials. As conjectured above, a few elements indeed grew prominently.

We considered that if a slight decay (forgetting) of value weights (c.f., [59-61]) was assumed, such a prominent growth of a few elements of w may be mitigated and alignment of w to c, beyond the initial loose alignment because of the non-negative constraint, may occur. These conjectures were indeed confirmed by simulations (Fig. 8Eb,c and Fig. 8F). The mean squared value error slightly increased when the value-weightdecay was assumed (Fig. 8G), however, presumably reflecting a decrease in developed values and a deterioration of learning because of the decay."

> *Rec7. I don't understand the qualitative difference between 4G and 4H. The difference seems to be smaller but there is still an apparent difference. Can this be quantified?*

We have added pointers indicating which were compared and statistical significance on Fig. 4D-H, and also Fig. 7 and Fig. 9C.

> *Rec8. More biologically realistic constraints.*

Are the weights allowed to become negative? - No.

Figure 6C - untrained RNN with non-negative x_i. Again - it was not explained what untrained RNN is. However, given my previous assumption, this is probably because the units developed in an untrained RNN is much further from representing a complete basis function. This cannot be done with only positive values. It would be useful to see network dynamics of units for untrained RNN. It might also be useful in all cases to estimate the dimensionality of the RNN. For 3 time-steps, it needs to be at least 3, and for more time steps as in Figure 4, larger.

As we mentioned in our reply to your public comment Pub3 (page 6-8), in the revised manuscript we examined models that incorporated inhibitory and excitatory units and followed Dale's law, which could still learn the tasks (Fig. 9, Line 479-520). We have also analyzed the dimensionality of network dynamics as we mentioned in our replies to your public comment Pub1 and recommendations Rec1 and Rec2.

> *Rec9. A new type of untrained RNN is introduced (Fig 6D) this is the first time an explanation of of the untrained RNN is given. Indeed, the dimensionality of the second type of untrained RNN should be similar to the bioVRNNrf. The results are still not good.*

In the model with the new type of untrained RNN whose elements were shuffled from trained bioVRNNrf, contribution ratios of later principal components of the trajectory of RNN activities (Fig. 6H gray dotted line) were indeed larger than those in the model with native untrained RNN (gray solid line) but still much smaller than those in the trained value RNN models with backprop (red line) or random feedback (blue line). It is considered that in value RNN, RNN connections were trained to realize high-dimensional trajectory, and shuffling did not generally preserve such an ability.

> *Rec10. The discussion is too long and verbose. This is not a review paper.*

We have made the original discussion much more compact (from 1686 words to 940 words). We have added new discussion, in response to the review comments, but the total length remains to be shorter than before (1589 words).

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *Tsurumi et al. show that recurrent neural networks can learn state and value representations in simple reinforcement learning tasks when trained with random feedback weights. The traditional method of learning for recurrent network in such tasks (backpropagation through time) requires feedback weights which are a transposed copy of the feed-forward weights, a biologically implausible assumption. This manuscript builds on previous work regarding "random feedback alignment" and "value-RNNs", and extends them to a reinforcement learning context. The authors also demonstrate that certain nonnegative constraints can enforce a "loose alignment" of feedback weights. The author's results suggest that random feedback may be a powerful tool of learning in biological networks, even in reinforcement learning tasks.*
>
> *Strengths:*
>
> *The authors describe well the issues regarding biologically plausible learning in recurrent networks and in reinforcement learning tasks. They take care to propose networks which might be implemented in biological systems and compare their proposed learning rules to those already existing in literature. Further, they use small networks on relatively simple tasks, which allows for easier intuition into the learning dynamics.*
>
> *Weaknesses:*
>
> *The principles discovered by the authors in these smaller networks are not applied to deeper networks or more complicated tasks, so it remains unclear to what degree these methods can scale up, or can be used more generally.*

We have examined extended models that incorporated inhibitory and excitatory units and followed Dale's law with certain assumptions, and found that these models could still learn the tasks. We have added these results in Fig. 9 and subsection “4.1 Models with excitatory and inhibitory units”.

We have also examined the performance of the models in a task in which distractor cue randomly appeared, finding that our models could still learn the state values much better than the models with untrained RNN. We have added these result in Fig. 10 and subsection “4.2 Task with distractor cue”.

Regarding the depth, we continue to think about it but have not yet come up with concrete ideas.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) I think the work would greatly benefit from more proofreading. There are language errors/oddities throughout the paper, I will list just a few examples from the introduction:*

Thank you for pointing this out. We have made revisions throughout the paper.

> *line 63: "simultaneously learnt in the downstream of RNN". Simultaneously learnt in networks downstream of the RNN? Simulatenously learn in a downstream RNN? The meaning is not clear in the original sentence.*

We have revised it to "simultaneously learnt in connections downstream of the RNN" (Line 67-68).

> *starting in line 65: " A major problem, among others.... value-encoding unit" is a run-on sentence and would more readable if split into multiple sentences.*

We have extensively revised this part, which now consists of short sentences (Line 70-75).

> *line 77: "in supervised learning of feed-forward network" should be either "in supervised learning of a feed-forward network" or "in supervised learning of feed-forward networks".*

We have changed "feed-forward network" to "feed-forward networks" (Line 83).

> *(2) Under what conditions can you use an online learning rule which only considers the influence of the previous timestep? It's not clear to me how your networks solve the temporal credit assignment problem when the cue-reward delay in your tasks is 3-5ish time steps. How far can you stretch this delay before your networks stop learning correctly because of this one-step assumption? Further, how much does feedback alignment constrain your ability to learn long timescales, such as in Murray, J.M. (2019)?*

The reason why our models can solve the temporal credit assignment problem at least to a certain extent is considered to be because temporal-difference (TD) learning, which we adopted, itself has a power to resolve temporal credit assignment, as exemplified in that TD(0) algorithms without eligibility trance can still learn the value of distant rewards. We have added a discussion on this in Line 702-705:

"…our models do not have "eligibility trace" (nor memorable/gated unit, different from the original value-RNN [26]), but could still solve temporal credit assignment to a certain extent because TD learning is by itself a solution for it (notably, recent work showed that combination of TD(0) and model-based RL well explained rat's choice and DA patterns [132])."

We have also examined the cases in which the cue-reward delay (originally 3 time steps) was elongated to 4, 5, or 6 time-steps, and our models with random feedback could still achieve better performance than the models with untrained RNN although the performance degraded as the cue-reward delay increased. We have added these results in Fig. 2M and Line 223-228 (for our original models without non-negative constraint)

"We further examined the cases with longer cue-reward delays. As shown in Fig. 2M, as the delay increased, the mean squared error of state values (at 3000-th trial) increased, but the relative superiority of oVRNNbp and oVRNNrf over the model with untrained RNN remained to hold, except for cases with small number of RNN units (5) and long delay (5 or 6) ($p < 0.0025$ in Wilcoxon rank sum test for oVRNNbp or oVRNNrf vs untrained for each number of RNN units for each delay)."

and Fig. 6J and Line 422-429 (for our extended models with non-negative constraint):

"Figure 6J shows the cases with longer cue-reward delays, with default or halved learning rates. As the delay increased, the mean squared error of state values (at 3000-th trial) increased, but the relative superiority of oVRNNbp-rev and oVRNNrf-bio over the models with untrained RNN remained to hold, except for a few cases with 5 RNN units (5 delay oVRNNrf-bio vs shuffled with default learning rate, 6 delay oVRNNrf-bio vs naive or shuffled with halved learning rate) ($p < 0.047$ in Wilcoxon rank sum test for oVRNNbp-rev or oVRNNrf-bio vs naive or shuffled untrained for each number of RNN units for each delay).”

As for the difficulty due to random feedback compared to backprop, there appeared to be little difference in the models without non-negative constraint (Fig. 2M), whereas in the models with nonnegative constraint, when the cue-reward delay was elongated to 6 time-steps, the model with random feedback performed worse than the model with backprop (Fig. 6J bottom-left panel).

> *(3) Line 150: Were the RNN methods trained with continuation between trials?*

Yes, we have added

“The oVRNN models, and the model with untrained RNN, were continuously trained across trials in each task, because we considered that it was ecologically more plausible than episodic training of separate trials.” in Line 147-150. This is considered to make learning of even the simple cue-reward association task nontrivial, as we describe in our reply to your comment 9 below.

> *(4) Figure 2I, J: indicate the statistical significance of the difference between the three methods for each of these measures.*

We have added statistical information for Fig. 2J (Line 198-203):

“As shown in the left panel of Fig. 2J, on average across simulations, oVRNNbp and oVRNNrf exhibited largely comparable performance and always outperformed the untrained RNN ($p < 0.00022$ in Wilcoxon rank sum test for oVRNNbp or oVRNNrf vs untrained for each number of RNN units), although oVRNNbp somewhat outperformed or underperformed oVRNNrf when the number of RNN units was small (≤10 ($p < 0.049$)) or large (≥25 ($p < 0.045$)), respectively.”

and also Fig. 6E (for non-negative models) (Line 385-390):

“As shown in the left panel of Fig. 6E, oVRNNbp-rev and oVRNNrf-bio exhibited largely comparable performance and always outperformed the models with untrained RNN ($p < 2.5\times10^{-12}$ in Wilcoxon rank sum test for oVRNNbp-rev or oVRNNrf-bio vs naive or shuffled untrained for each number of RNN units), although oVRNNbp-rev somewhat outperformed or underperformed oVRNNrf-bio when the number of RNN units was small (≤10 ($p < 0.00029$)) or large (≥25 ($p < 3.7\times10^{-6}$)), respectively…”

Fig. 2I shows distributions, whose means are plotted in Fig. 2J, and we did not add statistics to Fig. 2I itself.

> *(5) Line 178: Has learning reached a steady state after 1000 trials for each of these networks? Can you show a plot of error vs. trial number?*

We have added a plot of error vs trial number for original models (Fig. 2L, Line 221-223):

“We examined how learning proceeded across trials in the models with 20 RNN units. As shown in Fig. 2L, learning became largely converged by 1000-th trial, although slight improvement continued afterward.”

and non-negatively constrained models (Fig. 6I, Line 417-422):

“Figure 6I shows how learning proceeded across trials in the models with 20 RNN units. While oVRNNbp-rev and oVRNNrf-bio eventually reached a comparable level of errors, oVRNNrf-bio outperformed oVRNNbp-rev in early trials (at 200, 300, 400, or 500 trials; p <

0.049 in Wilcoxon rank sum test for each). This is presumably because the value weights did not develop well in early trials and so the backprop-type feedback, which was the same as the value weights, did not work well, while the non-negative fixed random feedback worked finely from the beginning."

As shown in these figures, learning became largely steady at 1000 trials, but still slightly continued, and we have added simulations with 3000 trials (Fig. 2M and Fig. 6J).

> *(6) Line 191: Put these regression values in the figure caption, as well as on the plot in Figure 3B.*

We have added the regression values in Fig. 3B and its caption.

> *(7) Line 199: This idea of being in the same quadrant is interesting, but I think the term "relatively close angle" is too vague. Is there another more quantatative way to describe this what you mean by this?*

We have revised this (Line 252-254) to "a vector that is in a relatively close angle with c , or more specifically, is in the same quadrant as (and thus within at maximum 90° from) c (for example, $[c_1\ c_2\ c_3]^T$ and $[0.5c_1\ 1.2c_2\ 0.8c_3]$T) "

> *(8) Line 275: I'd like to see this measure directly in a plot, along with the statistical significance.*

We have added pointers indicating which were compared and statistical significance on Fig. 4D-H, and also Fig. 7 and Fig. 9C.

> *(9) Line 280: Surely the untrained RNN should be able to solve the task if the reservoir is big enough, no? Maybe much bigger than 50 units, but still.*

We think this is not sure. A difficulty lies in that because we modeled the tasks in a continuous way rather than in an episodic way (as we mentioned in our reply to your comment 3), the activity of untrained RNN upon cue presentation should generally differ from trial to trial. Therefore, it was not trivial for RNN to know that cue presentation in different trials, even after random lengths of inter-trial interval, should constitute a same single state. We have added this note in Line 177-185:

"This inferiority of untrained RNN may sound odd because there were only four states from cue to reward while random RNN with enough units is expected to be able to represent many different states (c.f., [49]) and the effectiveness of training of only the readout weights has been shown in reservoir computing studies [50-53]. However, there was a difficulty stemming from the continuous training across trials (rather than episodic training of separate trials): the activity of untrained RNN upon cue presentation generally differed from trial to trial, and so it is non-trivial that cue presentation in different trials should be regarded as the same single state, even if it could eventually be dealt with at the readout level if the number of units increases."

The original value RNN study (Hennig et al., 2023, PLoS Comput Biol) also modeled tasks in a continuous way (though using BPTT for training) and their model with untrained RNN also showed considerably larger RPE error than the value RNN even when the number of RNN units was 100 (the maximum number plotted in their Fig. 6A).

> *(10) It's a bit confusing to compare Figure 4C to Figure 4D-H because there are also many features of D-H which do not match those of C (response to cue, response to late reward in task 1). It would make sense to address this in some way. Is there another way*

> *to calculate the true values of the states (e.g., maybe you only start from the time of the cue) which better approximates what the networks are doing?*

As we mentioned in our replies to your comments 3 and 9, our models with RNN were trained continuously across trials rather than separately for each episodic trial, and whether the models could still learn the state representation is a key issue. Therefore, starting learning from the time of cue would not be an appropriate way to compare the models, and instead we have made statistical comparison regarding key features, specifically, TD-RPEs at early and late rewards, as indicated in Fig. 4D-H.

> *(11) Line 309: Can you explain why this non-monotic feature exists? Why do you believe it would be more biologically plausible to assume monotonic dependence? It doesn't seem so straightforward to me, I can imagine that competing LTP/LTD mechanisms may produce plasticity which would have a non-monotic dependence on post-synaptic activity.*

Thank you for this insightful comment. As you suggested, non-monotonic dependence on the postsynaptic activity (BCM rule) has been proposed for unsupervised learning (cortical self-organization) (Bienenstock et al., 1982 J Neurosci), and there were suggestions that triplet-based STDP could be reduced to a BCM-like rule and additional components (Gjorgjieva et al., 2011 PNAS; Shouval, 2011 PNAS). However, the non-monotonicity appeared in our model, derived from the backprop rule, is maximized at the middle and thus opposite from the BCM rule, which is minimized at the middle (i.e., initially decrease and thereafter increase). Therefore we consider that such an increase-then-decreasetype non-monotonicity would be less plausible than a monotonic increase, which could approximate an extreme case (with a minimum dip) of the BCM rule. We have added a note on this point in Line 355-358:

"…the dependence on the post-synaptic activity was non-monotonic, maximized at the middle of the range of activity. It would be more biologically plausible to assume a monotonic increase (while an opposite shape of nonmonotonicity, once decrease and thereafter increase, called the BCM (Bienenstock-Cooper-Munro) rule has actually been suggested [56-58])."

> *(12) Line 363: This is the most exciting part of the paper (for me). I want to learn way more about this! Don't hide this in a few sentences. I want to know all about loose vs. feedback alignment. Show visualizations in 3D space of the idea of loose alignment (starting in the same quadrant), and compare it to how feedback alignment develops (ending in the same quadrant). Does this "loose" alignment idea give us an idea why the random feedback seems to settle at 45 degree angle? it just needs to get the signs right (same quadrant) for each element?*

In reply to this encouraging comment, we have made further analyses of the loose alignment. By the term "loose alignment", we meant that the value weight vector w and the feedback vector c are in the same (non-negative) quadrant, as you said. But what remained mysterious (to us) was while the angle between w and c was relatively close (loosely aligned) from the beginning, it appeared (as mentioned in the manuscript) that there was no further alignment over trials (and the angle actually settled at somewhat larger than 45°), despite that the same mechanism for feedback alignment that we derived for the model without non-negative constraint was expected to operate also under the nonnegative constraint. We have now clarified the reason for this, and found a way, introduction of slight decay (forgetting) of value weights, by which feedback alignment came to occur in the non-negatively constraint model. We have added this in Line 463-477:

"As mentioned above, while the angle between w and c was on average smaller than 90° from the beginning, there was no further alignment over trials. This seemed mysterious because the mechanism for feedback alignment that we derived for the models without non-negative constraint was expected to work also for the models with non-negative constraint. As a possible reason for the non-occurrence of feedback alignment, we guessed that one or a few element(s) of w grew prominently during learning, and so w became close to an edge or boundary of the non-negative quadrant and thereby angle between w and other vector became generally large (as illustrated in Fig. 8D). Figure 8Ea shows the mean±SEM of the elements of w ordered from the largest to smallest ones after 1500 trials. As conjectured above, a few elements indeed grew prominently.

We considered that if a slight decay (forgetting) of value weights (c.f., [59-61]) was assumed, such a prominent growth of a few elements of w may be mitigated and alignment of w to c, beyond the initial loose alignment because of the non-negative constraint, may occur. These conjectures were indeed confirmed by simulations (Fig. 8Eb,c and Fig. 8F). The mean squared value error slightly increased when the value-weightdecay was assumed (Fig. 8G), however, presumably reflecting a decrease in developed values and a deterioration of learning because of the decay."

As for visualization, because the model's dimension was high such as 12, we could not come up with better ways of visualization than the trial versus angle plot (Fig. 3A, 8A,F). Nevertheless, we would expect that the abovementioned additional analyses of loose alignment (with graphs) are useful to understand what are going on.

> *(13) Line 426: how does this compare to some of the reward modulated hebbian rules proposed in other RNNs? See Hoerzer, G. M., Legenstein, R., & Maass, W. (2014). Put another way, you arrived at this from a top-down approach (gradient descent->BP->approximated by RF->non-negativity constraint>leads to DA dependent modulation of Hebbian plasticity). How might this compare to a bottom up approach (i.e. starting from the principle of Hebbian learning, and adding in reward modulation)*

The study of Hoerzer et al. 2014 used a stochastic perturbation, which we did not assume but can potentially be integrated. On the other hand, Hoerzer et al. trained the readout of untrained RNN, whereas we trained both RNN and its readout. We have added discussion to compare our model with Hoerzer et al. and other works that also used perturbation methods, as well as other top-down approximation method, in Line 685-711 (reference 128 is Hoerzer et al. 2014 Cereb Cortex):

"As an alternative to backprop in hierarchical network, aside from feedback alignment [36], Associative Reward-Penalty (A_{R-P}) algorithm has been proposed [124-126]. In A_{R-P}, the hidden units behave stochastically, allowing the gradient to be estimated via stochastic sampling. Recent work [127] has proposed Phaseless Alignment Learning (PAL), in which high-frequency noise-induced learning of feedback projections proceeds simultaneously with learning of forward projections using the feedback in a lower frequency. Noise-induced learning of the weights on readout neurons from untrained RNN by reward-modulated Hebbian plasticity has also been demonstrated [128]. Such noise- or perturbation-based [40] mechanisms are biologically plausible because neurons and neural networks can exhibit noisy or chaotic behavior [129-131], and might improve the performance of value-RNN if implemented.

Regarding learning of RNN, "e-prop" [35] was proposed as a locally learnable online approximation of BPTT [27], which was used in the original value RNN 26. In e-prop, neuron-specific learning signal is combined with weight-specific locally-updatable "eligibility trace". Reward-based e-prop was also shown to work [35], both in a setup not introducing TD-RPE with symmetric or random feedback (their Supplementary Figure 5) and in another setup introducing TD-RPE with symmetric feedback (their Figure 4 and 5). Compared to these, our models differ in multiple ways.

First, we have shown that alignment to random feedback occurs in the models driven by TD-RPE. Second, our models do not have "eligibility trace" (nor memorable/gated unit, different from the original valueRNN [26]), but could still solve temporal credit assignment to a certain extent because TD learning is by itself a solution for it (notably, recent work showed that combination of TD(0) and model-based RL well explained rat's choice and DA patterns [132]). However, as mentioned before, single time-step in our models was assumed to correspond to hundreds of milliseconds, incorporating slow synaptic dynamics, whereas e-prop is an algorithm for spiking neuron models with a much finer time scale. From this aspect, our models could be seen as a coarsetime-scale approximation of e-prop. On top of these, our results point to a potential computational benefit of biological non-negative constraint, which could effectively limit the parameter space and promote learning."

Related to your latter point (and also replying to other reviewer's comment), we also examined the cases where the random feedback in our model was replaced with uniform feedback, which corresponds to a simple bottom-up reward-modulated triplet plasticity rule. As a result, the model with uniform feedback showed largely comparable, but somewhat worse, performance than the model with random feedback. We have added the results in Fig. 2J-right and Line 206-209 (for our original models without non-negative constraint):

"The green line in Fig. 2J-right shows the performance of a special case where the random feedback in oVRNNrf was fixed to the direction of $(1, 1, ..., 1)^T$ (i.e., uniform feedback) with a random coefficient, which was largely comparable to, but somewhat worse than, that for the general oVRNNrf (blue line)."

and Fig. 6E-right and Line 402-407 (for our extended models with non-negative constraint):

"The green and light blue lines in the right panels of Figure 6E and Figure 6F show the results for special cases where the random feedback in oVRNNrf-bio was fixed to the direction of $(1, 1, ..., 1)^T$ (i.e., uniform feedback) with a random non-negative magnitude (green line) or a fixed magnitude of 0.5 (light blue line). The performance of these special cases, especially the former (with random magnitude) was somewhat worse than that of oVRNNrf-bio, but still better than that of the models with untrained RNN. and also added a biological implication of the results in Line 644-652:

We have shown that oVRNNrf and oVRNNrf-bio could work even when the random feedback was uniform, i.e., fixed to the direction of $(1, 1, ..., 1)^T$, although the performance was somewhat worse. This is reasonable because uniform feedback can still encode scalar TD-RPE that drives our models, in contrast to a previous study [45], which considered DA's encoding of vector error and thus regarded uniform feedback as a negative control. If oVRNNrf/oVRNNrf-bio-like mechanism indeed operates in the brain and the feedback is near uniform, alignment of the value weights w to near (1, 1, ..., 1) is expected to occur. This means that states are (learned to be) represented in such a way that simple summation of cortical neuronal activity approximates value, thereby potentially explaining why value is often correlated with regional activation (fMRI BOLD signal) of cortical regions [113]."

> ***Reviewer #3 (Public review):***
>
> *Summary:*
>
> *The paper studies learning rules in a simple sigmoidal recurrent neural network setting. The recurrent network has a single layer of 10 to 40 units. It is first confirmed that feedback alignment (FA) can learn a value function in this setting. Then so-called bio-plausible constraints are added: (1) when value weights (readout) is non-negative, (2) when the activity is non-negative (normal sigmoid rather than downscaled between -0.5 and 0.5), (3) when the feedback weights are non-negative, (4) when the learning rule is*

> *revised to be monotic: the weights are not downregulated. In the simple task considered all four biological features do not appear to impair totally the learning.*
>
> *Strengths:*
>
> *(1) The learning rules are implemented in a low-level fashion of the form: (pre-synaptic-activity) x (post-synaptic-activity) x feedback x RPE. Which is therefore interpretable in terms of measurable quantities in the wet-lab.*
>
> *(2) I find that non-negative FA (FA with non negative c and w) is the most valuable theoretical insight of this paper: I understand why the alignment between w and c is automatically better at initialization.*
>
> *(3) The task choice is relevant since it connects with experimental settings of reward conditioning with possible plasticity measurements.*
>
> *Weaknesses:*
>
> *(4) The task is rather easy, so it's not clear that it really captures the computational gap that exists with FA (gradient-like learning) and simpler learning rule like a delta rule: RPE x (pre-synpatic) x (postsynaptic). To control if the task is not too trivial, I suggest adding a control where the vector c is constant c_i=1.*

We have examined the cases where the feedback was uniform, i.e., in the direction of (1, 1, ..., 1) in both models without and with non-negative constraint. In both models, the models with uniform feedback performed somewhat worse than the original models with random feedback, but still better than the models with untrained RNN. We have added the results in Fig. 2J-right and Line 206-209 (for our original models without non-negative constraint):

"The green line in Fig. 2J-right shows the performance of a special case where the random feedback in oVRNNrf was fixed to the direction of $(1, 1, ..., 1)^T$ (i.e., uniform feedback) with a random coefficient, which was largely comparable to, but somewhat worse than, that for the general oVRNNrf (blue line)."

and Fig. 6E-right and Line 402-407 (for our extended models with non-negative constraint):

"The green and light blue lines in the right panels of Figure 6E and Figure 6F show the results for special cases where the random feedback in oVRNNrf-bio was fixed to the direction of $(1, 1, ..., 1)^T$ (i.e., uniform feedback) with a random non-negative magnitude (green line) or a fixed magnitude of 0.5 (light blue line). The performance of these special cases, especially the former (with random magnitude) was somewhat worse than that of oVRNNrf-bio, but still better than that of the models with untrained RNN."

We have also added a discussion on the biological implication of the model with uniform feedback mentioned in our provisional reply in Line 644-652:

"We have shown that oVRNNrf and oVRNNrf-bio could work even when the random feedback was uniform, i.e., fixed to the direction of $(1, 1, ..., 1)^T$, although the performance was somewhat worse. This is reasonable because uniform feedback can still encode scalar TD-RPE that drives our models, in contrast to a previous study [45], which considered DA's encoding of vector error and thus regarded uniform feedback as a negative control. If oVRNNrf/oVRNNrf-bio-like mechanism indeed operates in the brain and the feedback is near uniform, alignment of the value weights w to near (1, 1, ..., 1) is expected to occur. This means that states are (learned to be) represented in such a way that simple summation of cortical neuronal activity approximates value, thereby potentially explaining why value is often correlated with regional activation (fMRI BOLD signal) of cortical regions [113]."

In addition, while preparing the revised manuscript, we found a recent simulation study, which showed that uniform feedback coupled with positive forward weights was effective in supervised learning of one-dimensional output in feed-forward network (Konishi et al., 2023, Front Neurosci).

We have briefly discussed this work in Line 653-655:

"Notably, uniform feedback coupled with positive forward weights was shown to be effective also in supervised learning of one-dimensional output in feed-forward network [114], and we guess that loose alignment may underlie it."

> *(5) Related to point 3), the main strength of this paper is to draw potential connection with experimental data. It would be good to highlight more concretely the prediction of the theory for experimental findings. (Ideally, what should be observed with non-negative FA that is not expected with FA or a delta rule (constant global feedback) ?).*

We have added a discussion on the prediction of our models, mentioned in our provisional reply, in Line 627-638:

"oVRNNrf predicts that the feedback vector c and the value-weight vector w become gradually aligned, while oVRNNrf-bio predicts that c and w are loosely aligned from the beginning. Element of c could be measured as the magnitude of pyramidal cell's response to DA stimulation. Element of w corresponding to a given pyramidal cell could be measured, if striatal neuron that receives input from that pyramidal cell can be identified (although technically demanding), as the magnitude of response of the striatal neuron to activation of the pyramidal cell. Then, the abovementioned predictions could be tested by (i) identify cortical, striatal, and VTA regions that are connected, (ii) identify pairs of cortical pyramidal cells and striatal neurons that are connected, (iii) measure the responses of identified pyramidal cells to DA stimulation, as well as the responses of identified striatal neurons to activation of the connected pyramidal cells, and (iv) test whether DA→pyramidal responses and pyramidal→striatal responses are associated across pyramidal cells, and whether such associations develop through learning."

Moreover, we have considered another (technically more doable) prediction of our model, and described it in Line 639-643:

"Testing this prediction, however, would be technically quite demanding, as mentioned above. An alternative way of testing our model is to manipulate the cortical DA feedback and see if it will cause (re-)alignment of value weights (i.e., cortical striatal strengths). Specifically, our model predicts that if DA projection to a particular cortical locus is silenced, effect of the activity of that locus on the value-encoding striatal activity will become diminished."

> *(6a) Random feedback with RNN in RL have been studied in the past, so it is maybe worth giving some insights how the results and the analyzes compare to this previous line of work (for instance in this paper [1]). For instance, I am not very surprised that FA also works for value prediction with TD error. It is also expected from the literature that the RL + RNN + FA setting would scale to tasks that are more complex than the conditioning problem proposed here, so is there a more specific take-home message about non-negative FA? or benefits from this simpler toy task? [1] https://www.nature.com/articles/s41467-020-17236-y*

As for a specific feature of non-negative models, we did not describe (actually did not well recognize) an intriguing result that the non-negative random feedback model performed generally better than the models without non-negative constraint with either backprop or random feedback (Fig. 2J-left versus Fig. 6E-left (please mind the difference in the vertical scales)). This suggests that the non-negative constraint effectively limited the parameter space and thereby learning became efficient. We have added this result in Line 392-395:

“Remarkably, oVRNNrf-bio generally achieved better performance than both oVRNNbp and oVRNNrf, which did not have the non-negative constraint (Wilcoxon rank sum test, vs oVRNNbp : p < 7.8×10,sup>−6 for 5 or ≥25 RNN units; vs oVRNNrf: $p < 0.021$ for ≤10 or ≥20 RNN units).”

Also, in the models with non-negative constraint, the model with random feedback learned more rapidly than the model with backprop although they eventually reached a comparable level of errors, at least in the case with 20 RNN units. This is presumably because the value weights did not develop well in early trials and so the backprop-based feedback, which was the same as the value weights, did not work well, while the non-negative fixed random feedback worked finely from the beginning. We have added this result in Fig. 6I and Line 417-422:

“Figure 6I shows how learning proceeded across trials in the models with 20 RNN units. While oVRNNbp-rev and oVRNNrf-bio eventually reached a comparable level of errors, oVRNNrf-bio outperformed oVRNNbp-rev in early trials (at 200, 300, 400, or 500 trials; $p < 0.049$ in Wilcoxon rank sum test for each). This is presumably because the value weights did not develop well in early trials and so the backprop-type feedback, which was the same as the value weights, did not work well, while the non-negative fixed random feedback worked finely from the beginning.”

We have also added a discussion on how our model can be positioned in relation to other models including the study you mentioned (e-prop by Bellec, ..., Maass, 2020) in subsection “Comparison to other algorithms” of the Discussion):

Regarding the slightly better performance of the non-negative model with random feedback than that of the non-negative model with backprop when the number of RNN units was large (mentioned in our provisional reply), state values in the backprop model appeared underdeveloped than those in the random feedback model. Slightly better performance of random feedback than backprop held also in our extended model incorporating excitatory and inhibitory units (Fig. 9B).

> *(6b) Related to task complexity, it is not clear to me if non-negative value and feedback weights would generally scale to harder tasks. If the task in so simple that a global RPE signal is sufficient to learn (see 4 and 5), then it could be good to extend the task to find a substantial gap between: global RPE, non-negative FA, FA, BP. For a well chosen task, I expect to see a performance gap between any pair of these four learning rules. In the context of the present paper, this would be particularly interesting to study the failure mode of non-negative FA and the cases where it does perform as well as FA.*

In the cue-reward association task with 3 time-steps delay, the non-negative model with random feedback performed largely comparably to the non-negative model with backprop, and this remained to hold in a task where distractor cue, which was not associated with reward, appeared in random timings. We have added the results in Fig. 10 and subsection “4.2 Task with distractor cue”.

We have also examined the cases where the cue-reward delay was elongated. In the case of longer cue-reward delay (6 time-steps), in the models without non-negative constraint, the model with random feedback performed comparably to (and slightly better than when the number of RNN units was large) the model with backprop (Fig. 2M). In contrast, in the models with non-negative constraint, the model with random feedback underperformed the model with backprop (Fig. 6J, left-bottom). This indicates a difference between the effect of non-negative random feedback and the effect of positive+negative random feedback.

We have further examined the performance of the models in terms of action selection, by extending the models to incorporate an actor-critic algorithm. In a task with inter-temporal choice (i.e., immediate small reward vs delayed large reward), the non-negative model with random feedback performed worse than the non-negative model with backprop when the number of RNN units was small. When the number of RNN increased, these models performed more comparably. These results are described in Fig. 11 and subsection “4.3 Incorporation of action selection”.

> *(7) I find that the writing could be improved, it mostly feels more technical and difficult than it should. Here are some recommendations:*
>
> *7a) for instance the technical description of the task (CSC) is not fully described and requires background knowledge from other paper which is not desirable.*
>
> *7b) Also the rationale for the added difficulty with the stochastic reward and new state is not well explained.*
>
> *7c) In the technical description of the results I find that the text dives into descriptive comments of the figures but high-level take home messages would be helpful to guide the reader. I got a bit lost, although I feel that there is probably a lot of depth in these paragraphs.*

As for 7a), 'CSC (complete serial compound)' was actually not the name of the task but the name of the 'punctate' state representation, in which each state (timing from cue) is represented in a punctate manner, i.e., by a one-hot vector such as (1, 0, ..., 0), (0, 1, ..., 0), ..., and (0, 0, ..., 1). As you pointed out, using the name of 'CSC' would make the text appearing more technical than it actually is, and so we have moved the reference to the name of 'CSC' to the Methods (Line 903-907):

“For the agents with punctate state representation, which is also referred to as the complete serial compound (CSC) representation [1, 48, 133], each timing from a cue in the tasks was represented by a 10-dimensional one-hot vector, starting from $(1\ 0\ 0\ \ldots\ 0)^T$ for the cue state, with the next state $(0\ 1\ 0\ \ldots\ 0)^T$ and so on.”

and in the Results we have instead added a clearer explanation (Line 163-165):

“First, for comparison, we examined traditional TD-RL agent with punctate state representation (without using the RNN), in which each state (time-step from a cue) was represented in a punctate manner, i.e., by a one-hot vector such as (1, 0, ..., 0), (0, 1, ..., 0), and so on.”

As for 7b), we have added the rationale for our examination of the tasks with probabilistic structures (Line 282-294):

“Previous work [54] examined the response of DA neurons in cue-reward association tasks in which reward timing was probabilistically determined (early in some trials but late in other trials). There were two tasks, which were largely similar but there was a key difference that reward was given in all the trials in one task whereas reward was omitted in some randomly determined trials in another task. Starkweather et al. [54] found that the DA response to later reward was smaller than the response to earlier reward in the former task, presumably reflecting the animal's belief that delayed reward will surely come, but the opposite was the case in the latter task, presumably because the animal suspected that reward was omitted in that trial. Starkweather et al.[54] then showed that such response patterns could be explained if DA encoded TD-RPE under particular state representations that incorporated the probabilistic structures of the task (called the 'belief state'). In that study, such state representations were 'handcrafted' by the authors, but the subsequent work [26] showed that the original value-RNN with backprop (BPTT) could develop similar representations and reproduce the experimentally observed DA patterns."

As for 7c), we have extensively revised the text of the results, adding high-level explanations while trying to reduce the lengthy low-level descriptions (e.g., Line 172-177 for Fig2E-G).

> *(8) Related to the writing issue and 5), I wished that "bio-plausibility" was not the only reason to study positive feedback and value weights. Is it possible to develop a bit more specifically what and why this positivity is interesting? Is there an expected finding with non-negative FA both in the model capability? or maybe there is a simpler and crisp take-home message to communicate the experimental predictions to the community would be useful?*

There is actually an unexpected finding with non-negative model: the non-negative random feedback model performed generally better than the models without non-negative constraint with either backprop or random feedback (Fig. 2J-left versus Fig. 6E-left), presumably because the nonnegative constraint effectively limited the parameter space and thereby learning became efficient, as we mentioned in our reply to your point 6a above (we did not well recognize this at the time of original submission).

Another potential merit of our present work is the simplicity of the model and the task. This simplicity enabled us to derive an intuitive explanation on why feedback alignment could occur. Such an intuitive explanation was lacking in previous studies while more precise mathematical explanations did exist. Related to the mechanism of feedback alignment, one thing remained mysterious to us at the time of original submission. Specifically, in the non-negatively constraint random feedback model, while the angle between the value weight (w) and the random feedback (c) was relatively close (loosely aligned) from the beginning, it appeared (as mentioned in the manuscript) that there was no further alignment over trials (and the angle actually settled at somewhat larger than 45°), despite that the same mechanism for feedback alignment that we derived for the model without non-negative constraint was expected to operate also under the non-negative constraint. We have now clarified the reason for this, and found a way, introduction of slight decay (forgetting) of value weights, by which feedback alignment came to occur in the non-negatively constraint model. We have added this in Line 463-477:

"As mentioned above, while the angle between w and c was on average smaller than 90° from the beginning, there was no further alignment over trials. This seemed mysterious because the mechanism for feedback alignment that we derived for the models without non-negative constraint was expected to work also for the models with non-negative constraint. As a possible reason for the non-occurrence of feedback alignment, we guessed that one or a few element(s) of w grew prominently during learning, and so w became close to an edge or boundary of the non-negative quadrant and thereby angle between w and other vector became generally large (as illustrated in Fig. 8D). Figure 8Ea shows the mean±SEM of the elements of w ordered from the largest to smallest ones after 1500 trials. As conjectured above, a few elements indeed grew prominently.

We considered that if a slight decay (forgetting) of value weights (c.f., [59-61]) was assumed, such a prominent growth of a few elements of w may be mitigated and alignment of w to c, beyond the initial loose alignment because of the non-negative constraint, may occur. These conjectures were indeed confirmed by simulations (Fig. 8Eb,c and Fig. 8F). The mean squared value error slightly increased when the value-weightdecay was assumed (Fig. 8G), however, presumably reflecting a decrease in developed values and a deterioration of learning because of the decay."

Correction of an error in the original manuscript

In addition to revising the manuscript according to your comments, we have made a correction on the way of estimating the true state values. Specifically, in the original manuscript, we defined states by relative time-steps from a reward and estimated their values by calculating the sums of discounted future rewards starting from them through simulations. However, we assumed variable inter-trial intervals (ITIs) (4, 5, 6, or 7 time-steps with equal probabilities), and so until receiving cue information, agent should not know when the next reward will come. Therefore, states for the timings up to the cue timing cannot be defined by the upcoming reward, but previously we did so (e.g., state of "one timestep before cue") without taking into account the ITI variability.

We have now corrected this issue, having defined the states of timings with respect to the previous (rather than upcoming) reward. For example, when ITI was 4 time-steps and agent existed in its last time-step, agent will in fact receive a cue at the next time-step, but agent should not know it until actually receiving the cue information and instead should assume that s/he was at the last time-step of ITI (if ITI was 4), last − 1 (if ITI was 5), last − 2 (if ITI was 6), or last − 3 (if ITI was 7) with equal probabilities (in a similar fashion to what we considered when thinking about state definition for the probabilistic tasks). We estimated the true values of states defined in this way through simulations. As a result, the corrected true value of the cue-timing has become slightly smaller than the value described in the original manuscript (reflecting the uncertainty about ITI length), and consequently small positive TD-RPE has now appeared at the cue timing.

Because we measured the performance of the models by squared errors in state values, this correction affected the results reporting the performance. Fortunately, the effects were relatively minor and did not largely alter the results of performance comparisons. However, we sincerely apologize for this error. In the revised manuscript, we have used the corrected true values throughout the manuscript, and we have described the ways of estimating these values in Line 919-976.

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